`

MITIGATING HALLUCINATIONS IN LARGE LANGUAGE MODELS USING RETRIEVAL-AUGMENTED GENERATION

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Final Thesis Report

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DEDICATION

To my parents, HFO. Satish Kumar Sharma and Daizy Sharma,

who instilled in me the belief that education is the most enduring investment one can make in oneself, and whose quiet encouragement sustained me through the most demanding phases of this journey.

To my wife Priyanka Sharma,

for your endless patience, love, and the sacrifices that allowed me to see this through.

And to every practitioner and researcher working to make artificial intelligence more trustworthy, more grounded, and more worthy of the responsibility placed upon it this work is offered in that spirit.

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I am also grateful to the creators of FinanceBench (Islam et al., 2023) for releasing a high-quality, publicly available benchmark for financial question answering. Without a rigorous, domain-specific evaluation corpus, this study's conclusions would have rested on far weaker empirical ground.

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ABSTRACT

Title: Mitigating Hallucinations in Large Language Models Using Retrieval-Augmented Generation

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Background. Large Language Models (LLMs) have been adopted rapidly across the Banking, Financial Services, and Insurance (BFSI) sector for compliance query resolution, regulatory document analysis, and policy question answering. Their principal deployment risk of hallucination, the generation of factually unsupported content, carries severe consequences in regulated environments, including regulatory liability and erosion of institutional trust. Retrieval-Augmented Generation (RAG), which grounds LLM generation in passages retrieved from authoritative document corpora at inference time, is the most widely adopted architectural mitigation. However, the optimal RAG pipeline configuration for BFSI document QA encompassing retrieval strategy, chunking strategy, retrieval depth, and citation grounding has not previously been subject to controlled factorial empirical investigation.

Objectives. This study conducts the first factorial empirical comparison of dense vector retrieval, sparse BM25 retrieval, and hybrid Reciprocal Rank Fusion (RRF) retrieval on a BFSI-specific benchmark, measuring hallucination rate and faithfulness as primary outcomes alongside context precision, answer F1, and per-query token cost. Four co-varying pipeline design variables are investigated: retrieval strategy (3 levels), chunking strategy (4 levels: fixed-256, fixed-512, semantic, sentence-window), retrieval depth (top-k $\in \{1, 3, 5, 10\}$), and citation grounding (enabled / disabled).

Methods. A modular Python RAG pipeline was implemented using LangChain and LlamaIndex, with LLaMA 3.2-3B as the fixed generation backbone (temperature = 0) and qwen3-embedding:4b as the fixed embedding model. Dense indexing was performed using TurboQuant (Zandieh et al., 2026). The evaluation benchmark was FinanceBench (Islam et al., 2023), comprising 150 questions grounded in real SEC corporate filings. A full factorial design yielded 96 unique pipeline configurations and 14,400 query-level observations. Hallucination rate was measured using the DeepEval G-Eval framework; faithfulness and retrieval quality metrics were measured using RAGAS. Statistical analysis employed the Kruskal-Wallis H-test for main effects, aligned rank transform ANOVA for interaction effects, and the paired Wilcoxon signed-rank test for citation grounding comparisons.

Results. Hybrid RRF retrieval achieved the lowest mean hallucination rate (0.441), significantly outperforming dense retrieval (0.518) and sparse BM25 (0.503) ($H = 18.722$, $p < 0.001$). No significant effect of retrieval strategy on faithfulness was detected ($H = 2.759$, $p = 0.252$). Chunking strategy was the dominant determinant of context precision ($\eta^2 p = 0.915$), with semantic chunking achieving the highest precision across all retrieval configurations. A significant Retrieval × Chunking interaction was detected ($F(6,84) = 2.28$, $p = 0.043$, $\eta^2 p = 0.140$). Retrieval depth yielded meaningful hallucination reductions up to top-10 ($\Delta = +0.040$ from top-5 to top-10), contradicting the commonly assumed top-5 saturation heuristic. Prompt-level citation grounding produced a statistically significant faithfulness improvement of +0.023 ($W = 34.000$, $p < 0.0001$) across all pipeline configurations. The optimal configuration R3_hybrid, C2_fixed_512, top-5, G1_enabled achieved a hallucination rate of 0.34 and faithfulness of 0.985 at a mean token cost of 4,655 per query.

Conclusions. Hybrid retrieval is the recommended default for BFSI RAG deployments. Chunking strategy selection deserves at least as much engineering attention as retrieval algorithm selection, with semantic or 512-token fixed-size chunking preferred over sentence-window chunking. Increasing retrieval depth to top-10 remains worthwhile where token cost permits. Prompt-level citation grounding is a zero-cost faithfulness improvement that should be universally adopted. These findings provide the first domain-specific, factorial, multi-metric evidence base to guide RAG pipeline design decisions in BFSI policy question answering environments.

Keywords: *Retrieval-Augmented Generation, Large Language Models, Hallucination Mitigation, BFSI, Financial NLP, Dense Retrieval, BM25, Hybrid Retrieval, Chunking Strategy, Citation Grounding, FinanceBench, RAGAS, Faithfulness, Factorial Experiment*

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TABLE OF CONTENTS

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LIST OF TABLES

`

LIST OF FIGURES

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LIST OF ABBREVIATIONS

BERT ...………………………….. Bidirectional Encoder Representations from Transformers

GPT ...…………………………………………………… Generative Pre-trained Transformer

FATCA……………………………………………... Foreign Account Tax Compliance Act

NLP ...………………………………………………………. Natural Language Processing

HyDE ……………………………………………….. Hypothetical Document Embeddings

RAGAS ………………………………… Retrieval-Augmented Generation Assessment Suite

DPR ……………………………………………………………… Dense Passage Retrieval

FAISS ………………………………………………………... Facebook AI Similarity Search

RRF ……………………………………………………………….. Reciprocal Rank Fusion

TREC ……………………………………………………………... Text Retrieval Conference

DORA …………………………………………………… Digital Operational Resilience Act

FFIEC …………………………………... Federal Financial Institutions Examination Council

AGREE ……………………………………………….. Adaptation for Grounding Enhancement

NLI ………………………………………………………… Natural Language Inference

ESMA ……………………………………... European Securities and Markets Authority

MTEB ………………………………………………Massive Text Embedding Benchmark

SEC ………………………………………………… Securities and Exchange Commission

FCA……………………………………………………………Financial Conduct Authority

EBA………………………………………………………… European Banking Authority

OCC ……………………………………………Office of the Comptroller of the Currency

AML……………………………………….............................Anti-Money-Laundering

MMR……………………………………………………Maximal Marginal Relevance

QJL……………………………………………………Quantized Johnson-Lindenstrauss

EM………………………………………………………………………Exact Match

ART…………………………………………………………..Aligned Rank Transform

HR……………………………………………………………………..Hallucination Rate

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CHAPTER 1
INTRODUCTION

1.1 Background of the Study

The financial services industry operates at the intersection of extraordinary complexity and extraordinary consequence. Every day, institutions spanning retail banks, insurance underwriters, asset managers, and central clearinghouses process billions of transactions, interpret hundreds of regulatory obligations, serve millions of customers, and bear fiduciary and legal duties that extend across multiple jurisdictions. The documentation that governs this activity capital adequacy frameworks, anti-money-laundering policies, underwriting guidelines, compliance manuals, and prudential standards runs to millions of pages, is continuously amended by regulatory bodies such as the Basel Committee on Banking Supervision, the Financial Conduct Authority (FCA), the European Banking Authority (EBA), and the United States Federal Reserve, and demands a level of precision in its interpretation that leaves no practical margin for ambiguity.

It is against this backdrop that the emergence of Large Language Models (LLMs) as a commercial technology has generated both genuine excitement and legitimate concern. The intellectual journey that led to modern LLMs began with the introduction of the Transformer architecture by (Vaswani et al., 2017), a design built entirely on self-attention mechanisms that replaced the recurrent and convolutional structures of prior neural networks with a parallelisable, scalable framework for sequence modelling. This innovation proved transformative. Pre-trained language models such as BERT (Devlin et al., 2019) demonstrated that deep bidirectional representations of text, learned from vast unlabelled corpora, could be fine-tuned to achieve state-of-the-art performance across diverse language understanding tasks. The subsequent scaling of these architectures most prominently in the GPT-3 paper of (Brown et al., 2020), which demonstrated that language models with 175 billion parameters could perform competently on tasks they had never been explicitly trained to perform established that scale itself was a mechanism for emergent capability. Open-weight successors, including the LLaMA family (Touvron et al., 2023), made research-grade LLMs accessible to academic and industrial communities without dependency on proprietary APIs, democratising the capacity to build and evaluate LLM-based applications.

The BFSI sector has been among the fastest adopters of LLM capabilities, and the commercial rationale is straightforward. An intelligent system capable of answering a compliance officer's question about Basel III capital buffer requirements in seconds, summarising a forty-page

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insurance policy for a customer service agent, or flagging inconsistencies between a draft contract and regulatory obligations represents a genuine operational advantage. A 2024 Deloitte industry survey estimated that financial institutions deploying AI-assisted compliance tools reduced the cost of regulatory change management by twenty-three percent on average, with time-to-resolution for compliance queries falling by over forty percent. These efficiency gains are not hypothetical, they are already being realised and competitive pressure is accelerating their adoption across the sector.

Yet beneath this promising trajectory lies a structural problem that the field has not solved and cannot afford to ignore. LLMs are generative systems, they produce text by learning statistical patterns over training data and generating sequences that are locally plausible given what has come before. They do not look up facts; they reconstruct them from probability distributions encoded in their parameters during pre-training. This mechanism, which is responsible for their remarkable fluency and flexibility, is also responsible for their most consequential failure mode called hallucination. When an LLM encounters a query that touches on knowledge it does not reliably possess, or that requires precise factual accuracy beyond what statistical pattern-matching can guarantee, it does not acknowledge uncertainty. It generates a response that is grammatically polished, contextually plausible, and factually wrong. In a general consumer setting, a hallucinated summary of a film plot or a misattributed historical quote is at most an inconvenience. In BFSI environments, a hallucinated capital requirement figure, a misquoted regulatory clause, or an invented risk threshold can trigger regulatory action, generate legal liability, and erode the institutional trust that financial services depend on.

The scale of this problem is not speculative. (Joshi, 2025) estimates that enterprise losses attributable to hallucination-related LLM incidents in BFSI contexts exceed two hundred and fifty million US dollars annually. (Kang and Liu, 2023) provide empirical evidence that LLMs produce financially erroneous outputs at rates substantially higher on financial reasoning tasks than on general-domain benchmarks, particularly for questions involving numerical precision, temporal specificity, and regulatory cross-referencing, precisely the question types most common in compliance QA. (Tonmoy et al., 2024) surveying over thirty-two mitigation approaches in the academic literature, document four principal hallucination categories *i.e.*: factual contradiction, contextual irrelevance, logical inconsistency and temporal disorientation The survey noted that no single mitigation technique has demonstrated consistent, reliable suppression across all four. The documented consequences in real deployments are severe: in one case, an internal chatbot at a global financial institution generated a fictitious provision under the Foreign Account Tax Compliance Act (FATCA) that was acted upon by compliance

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staff before detection in audit; in another, an LLM-generated investment summary misstated the credit rating of a debt instrument, requiring legal intervention to correct.

The architectural response that has attracted the greatest academic and commercial attention is Retrieval-Augmented Generation (RAG), introduced in its modern form by (Lewis et al., 2021). The core insight of RAG is simple but powerful: rather than requiring the language model to recall information entirely from its parametric memory, a retrieval component fetches the relevant passages from an authoritative document corpus at inference time and supplies them to the model as grounded context for generation. The model's role shifts from recall to reasoning. It must interpret, synthesise, and respond to what it has been shown, rather than what it remembers. In principle, this architectural change should substantially reduce hallucination rates. If the correct answer is present in the retrieved context and the model generates faithfully from that context, the probability of producing a factually incorrect response drops dramatically. In practice, however, the performance of a RAG system is determined by a cascade of design choices, how documents are ingested and segmented, how the retrieval index is constructed, which retrieval algorithm is used to select relevant passages, how many passages are retrieved and how the generation prompt is structured. The evidence that any particular combination of these choices is optimal for BFSI policy question answering is, at the outset of this study, absent.

Three retrieval paradigms dominate the current landscape. Sparse retrieval, represented principally by BM25 (Robertson and Zaragoza, 2009), operates on exact lexical matching, scoring documents by a weighted combination of term frequency and inverse document frequency. Dense retrieval, pioneered by the Dense Passage Retrieval (DPR) system of (Karpukhin et al., 2020) and refined through successive generations of embedding models including Sentence-BERT (Reimers and Gurevych, 2019) and modern instruction-tuned encoders such as qwen3-embedding:4b (Zhang et al., 2025), embeds queries and documents into shared semantic vector spaces and retrieves by approximate nearest-neighbour search. Hybrid retrieval fuses ranked outputs from both paradigms the most commonly through Reciprocal Rank Fusion (Cormack et al., 2009) combining the terminological precision of sparse methods with the semantic coverage of dense methods without requiring score normalisation across incompatible scales. Each paradigm has distinct theoretical strengths; which is most effective for the specific characteristics of BFSI regulatory text, with its precise terminology, numerical exactness requirements, and dense cross-referencing is an empirical question that this study sets out to answer.

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Vector indexing underlies the dense retrieval pathway and introduces its own design considerations. The adoption of TurboQuant (Zandieh et al., 2026), accepted at ICLR 2026, as the indexing layer in this study represents a deliberate choice of a state-of-the-art quantization algorithm over conventional approximate nearest-neighbour libraries. TurboQuant achieves near-optimal distortion rates through a two-stage data-oblivious quantization scheme that requires no offline calibration and reduces indexing time to near zero while maintaining retrieval accuracy comparable to full-precision baselines. A combination of properties particularly suited to the modular, reproducible pipeline design that this study requires.

Beyond retrieval strategy, two further design choices emerge from the literature as consequential but underexplored in the BFSI context. Document chunking the segmentation of source documents into retrievable units determines the granularity at which the retrieval system operates, mediating a fundamental trade-off between retrieval precision and generation-stage context availability (Smith and Troynikov, 2024; Hladěna et al., 2025). Citation grounding the explicit instruction to the generation model to attribute every factual claim in its response to a specific retrieved passage, represents a generation-stage constraint that may substantially improve faithfulness by preventing the model from supplementing retrieved evidence with parametric confabulation (Maynez et al., 2020; Ye et al., 2024). Both of these variables are included as controlled experimental factors in this study, making it to the best of the author's knowledge. The first factorial investigation of retrieval strategy, chunking strategy, retrieval depth, and citation grounding as co-varying design choices in a BFSI QA setting.

The evaluation benchmark is FinanceBench (Islam et al., 2023), a purpose-built financial question answering dataset constructed from real SEC corporate filings 10-K annual reports, 10-Q quarterly filings, and earnings call transcripts with annotated question-answer pairs that require both factual extraction and numerical reasoning. FinanceBench's grounded structure, authentic document provenance, and coverage of the multi-section and cross-document question types that characterise real compliance QA tasks makes it the most appropriate evaluation corpus available for this study. The primary evaluation tools are the RAGAS framework (Es et al., 2024) for automated faithfulness and retrieval quality scoring, and the DeepEval framework (AI, 2023) implementing G-Eval (Liu et al., 2023) for hallucination rate measurement and broader generation quality assessment.

This study is therefore positioned at the intersection of four active research streams *i.e.* LLM hallucination mitigation, RAG pipeline engineering, information retrieval in specialised domains and automated evaluation of grounded generation and applies their methods to a domain of substantial economic and societal importance.

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1.2 Problem Statement

Despite the rapid growth of LLM deployments across financial services and the demonstrated capability of RAG architectures to reduce factual errors in general-domain settings, a specific and consequential problem remains unresolved: there is no published, controlled, empirical comparison of dense, sparse, and hybrid retrieval strategies evaluated on a financial domain benchmark and measured against hallucination rate and faithfulness as primary outcomes that also accounts for the design choices of chunking strategy, retrieval depth, and citation grounding as co-varying experimental factors.

This gap is not merely an academic oversight. Financial institutions evaluating RAG architectures for production deployment face a practical decision problem: they must choose a retrieval strategy, a chunking configuration, a retrieval depth, and a prompting approach without access to domain-specific evidence on which combination of choices best serves their reliability requirements. The evidence that does exist is fragmented in three ways.

The first fragmentation is domain specificity. The majority of controlled RAG comparisons in the academic literature are conducted on general-domain corpora such as Wikipedia-derived open QA benchmarks or news article collections. These corpora have fundamentally different characteristics from financial regulatory text: they are less terminologically precise, less numerically dense, more rhetorically varied, and less structured around cross-referenced provisions. The retrieval performance rankings observed on general-domain corpora may not transfer to BFSI settings, and the provocative recent finding of (Akarsu et al., 2026) that BM25 outperforms dense retrieval on financial document corpora provides direct empirical grounds for scepticism about such transfer. (Setty et al., 2024) and (Kim et al., 2025) both investigate retrieval in financial QA contexts but neither provides a full three-way comparison of dense, sparse, and hybrid approaches with hallucination rate as a primary measured outcome.

The second fragmentation is variable isolation. Published studies that do examine retrieval strategy on financial text tend to hold chunking strategy constant or select it as a preprocessing step rather than treating it as an experimental variable. This is a significant methodological limitation: chunking strategy determines the granularity of the units the retrieval system operates on and has been shown to materially affect retrieval precision through mechanisms distinct from those of the retrieval algorithm itself (Smith and Troynikov, 2024; Hladěna et al., 2025). The interaction between retrieval strategy and chunking strategy whether, for instance, BM25's lexical matching advantage on regulatory text is preserved or diminished under different chunk granularity conditions cannot be assessed from studies that vary only one of these factors.

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The third fragmentation is metric coverage. Existing studies in the BFSI RAG literature typically report a subset of the performance dimensions relevant to deployment decisions. Some report retrieval precision but not generation faithfulness; others report faithfulness but not latency or token cost; none simultaneously characterises hallucination rate, faithfulness, retrieval precision, response latency, and per-query token cost across a factorial matrix of pipeline configurations. In production financial systems, deployment decisions involve trade-offs across all of these dimensions: a configuration that minimises hallucination at the cost of a response latency of ten seconds may be unacceptable for customer-facing applications while remaining entirely suitable for overnight batch compliance reporting. Without a simultaneous multi-dimensional profile, practitioners cannot make evidenced trade-off decisions.

Beyond these empirical gaps, a conceptual gap also exists around citation grounding. The AGREE framework of (Ye et al., 2024) demonstrates that fine-tuned citation grounding substantially improves faithfulness, but fine-tuning is not available in most enterprise RAG deployments, which operate LLMs in inference-only mode. Whether prompt-level citation grounding instructions without any model training produce a measurable and consistent faithfulness improvement across different retrieval strategy and chunking conditions has not been empirically investigated in the published literature. This is a practically important question: if prompt-level citation grounding is effective, it represents a zero-cost improvement available to any RAG deployment; if it is ineffective, practitioners should not invest engineering effort in implementing it.

Taken together, these gaps define the problem this study addresses: the absence of a controlled, factorial, multi-metric empirical investigation of RAG pipeline design choices for BFSI policy question answering. The study is designed specifically to fill this gap, not to build the most sophisticated possible RAG system, but to produce reliable, comparative, domain-specific evidence that can inform practical deployment decisions.

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1.3 Aim and Objectives

The aim of this study is to design and conduct a systematic empirical evaluation of Retrieval-Augmented Generation pipeline configurations that identifies which combinations of retrieval strategy, chunking approach, retrieval depth, and citation grounding most effectively reduce hallucinations in BFSI policy question answering, while characterising the associated trade-offs in faithfulness, accuracy, response latency, and per-query token cost.

In pursuit of this aim, the study has five specific objectives, listed in order of importance:

Objective 1: To conduct a critical and comprehensive review of the existing academic literature on LLM hallucinations and RAG-based mitigation approaches, with particular emphasis on empirical studies involving financial and regulatory domain corpora, in order to establish the theoretical foundations of the study, identify the design variables of interest, and precisely locate the research gaps that this study is positioned to address.

Objective 2: To implement a modular, reproducible Python-based RAG pipeline that supports interchangeable retrieval strategies (dense, sparse, and hybrid), chunking schemes (fixed-size at 256 and 512 tokens, semantic, and sentence-window), retrieval depths (top-k $\in \{1, 3, 5, 10\}$), and citation-grounding prompt configurations (enabled and disabled), using the FinanceBench dataset as the primary evaluation corpus, with LLaMA 3.2-3B as the fixed generation backbone and qwen3-embedding:4b as the fixed embedding model.

Objective 3: To conduct a controlled factorial experiment across all pipeline configurations, systematically measuring hallucination rate, faithfulness score, answer accuracy, response latency, and per-query token cost, in order to identify the performance effects of each design variable independently and in combination.

Objective 4: To produce a ranked set of RAG pipeline configurations, backed by statistical analysis, that characterises the accuracy-latency, accuracy-cost, and faithfulness-cost trade-offs across the experimental conditions, providing actionable guidance for practitioners responsible for deploying RAG systems in BFSI environments.

Objective 5: To release the complete codebase, evaluation scripts, experimental configuration files, and a reproducible run log as an open-source repository, ensuring that the study's findings can be independently verified, extended to additional corpora or retrieval methods, and built upon by the research community.

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1.4 Research Questions

The study is organised around one primary research question and four supporting sub-questions. The primary question defines the central empirical focus; the sub-questions decompose it into testable hypotheses corresponding to the four independent variables in the experimental design.

Primary Research Question: How do retrieval strategy, chunking approach, retrieval depth, and citation grounding each affect, and interact to jointly affect, hallucination rate and faithfulness in a RAG-based question answering pipeline operating on BFSI policy documents?

RQ1: Which retrieval strategy: dense vector search, sparse BM25 keyword retrieval, or hybrid Reciprocal Rank Fusion produces the lowest hallucination rate and highest faithfulness score when applied to BFSI policy question answering on the FinanceBench benchmark, and how do the strategies compare across all five evaluation metrics?

This question addresses the central retrieval variable. It is motivated by the contrasting theoretical strengths of each strategy lexical precision for sparse methods, semantic coverage for dense methods, and complementary fusion for hybrid methods and by the documented but unresolved debate in the literature about which approach best suits the characteristics of financial regulatory text. The comparison is made under identical chunking, retrieval depth, and prompting conditions to ensure the effect is attributable to retrieval strategy alone.

RQ2: How does chunking strategy specifically the choice between fixed-size token chunking at 256 and 512 tokens, semantic chunking, and sentence-window chunking affect retrieval precision and factual correctness in BFSI policy QA, and does chunking strategy interact with retrieval strategy in its effect on performance outcomes?

This question addresses the document segmentation variable and is motivated by the demonstrated sensitivity of retrieval quality to chunk granularity and boundary placement (Smith and Troynikov, 2024; Hladěna et al., 2025). It extends prior work by examining chunking not in isolation but in combination with retrieval strategy, enabling the detection of interaction effects that single-variable studies cannot identify.

RQ3: What is the relationship between retrieval depth the number of top-k passages retrieved per query and hallucination rate across the different retrieval strategies and chunking configurations, and at what retrieval depth do diminishing returns in hallucination reduction become apparent?

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This question addresses the retrieval depth variable and is motivated by the expectation that increasing k improves recall at the cost of injecting noise into the generation context. The point of diminishing returns and the position of each retrieval strategy on the recall-noise curve is of direct practical relevance: it determines how many retrieved passages a production system should include in its generation prompt to maximise accuracy per unit of token cost.

RQ4: Does the addition of an explicit citation-grounding instruction in the generation prompt requiring the model to attribute every factual claim to a specific retrieved passage produce a statistically significant improvement in faithfulness score on FinanceBench, and is this effect consistent across retrieval strategy and chunking configurations?

This question addresses the citation grounding variable and is motivated by the absence of any controlled investigation of prompt-level citation grounding in the BFSI context. Unlike the AGREE approach of (Ye et al., 2024), which requires model fine-tuning, this study examines whether a zero-cost prompt instruction is sufficient to produce measurable faithfulness gains, which would have immediate practical implications for any financial institution operating an LLM in inference-only mode.

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1.5 Scope of the Study

A clearly defined scope is essential for any empirical study that must produce interpretable, reliable results within defined resource and time constraints. This section documents what this study covers and what it intentionally excludes, with the reasoning for each boundary.

Within the scope of this study are the following elements. The primary domain is BFSI policy question answering specifically, the task of answering factual and numerical questions grounded in real corporate financial filings, as represented in the FinanceBench dataset (Islam et al., 2023), which provides authentic SEC 10-K annual reports, 10-Q quarterly filings, and earnings call transcripts from publicly traded companies alongside annotated question-answer pairs. Three retrieval strategies are compared: dense vector search using TurboQuant-indexed embeddings generated by the qwen3-embedding:4b model (Zhang et al., 2025), sparse keyword retrieval using BM25 via the Rank-BM25 Python library (Robertson and Zaragoza, 2009), and hybrid fusion via Reciprocal Rank Fusion (Cormack et al., 2009). Four chunking strategies are evaluated: fixed-size chunking at 256 tokens, fixed-size chunking at 512 tokens, semantic chunking via LlamaIndex's SemanticSplitterNodeParser, and sentence-window chunking with a window of plus or minus two sentences around the matched passage. Retrieval depth is varied across four levels: top-1, top-3, top-5, and top-10 retrieved passages. Citation grounding is evaluated as a binary variable: enabled (a prompt instruction requiring each factual claim to be tagged with its source passage identifier) and disabled (a standard RAG prompt without citation attribution requirements). The LLM backbone used for all generation is LLaMA 3.2-3B (Touvron et al., 2023) operated at inference temperature zero for deterministic outputs. Evaluation is performed using the RAGAS framework (Es et al., 2024) for faithfulness and retrieval quality metrics, and the DeepEval framework (AI, 2023) implementing G-Eval (Liu et al., 2023) for hallucination rate measurement.

Beyond the scope of this study are the following elements, each excluded for clearly stated reasons. Graph-augmented retrieval approaches, including GraphRAG and similar knowledge-graph-based architectures (Barry et al., 2025), are acknowledged in the literature review but excluded from the empirical comparison. Their additional pre-processing overhead graph construction, entity resolution, and relationship extraction constitutes a structurally distinct engineering challenge that would confound a controlled comparison of retrieval strategies operating on the same document representation. Graph-based RAG is best evaluated in a dedicated study designed around its specific design variables. Real-time or streaming RAG systems, operating on live data feeds such as market data, news wires, or continuously updated regulatory repositories, are excluded because FinanceBench provides a static, curated corpus

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that enables controlled comparison; extending to dynamic corpora would introduce data freshness and synchronisation variables that are outside the study's empirical scope. Training and fine-tuning of any model component including the LLM backbone, the embedding model, and any re-ranking component are excluded. All models operate in inference mode only. This constraint is deliberate: it ensures that the study's findings apply to the broadest class of production RAG deployment, where fine-tuning is typically unavailable due to infrastructure, cost, or regulatory restrictions on model modification. Multimodal inputs, including document images, tables rendered as images, charts, and visual financial instruments, are excluded because FinanceBench focuses on textual content, and multimodal processing would require vision-language model architectures that are outside the study's scope. Healthcare, legal, and other regulated domain corpora are not included; the domain-specific characteristics of BFSI regulatory text its terminological conventions, document structure, and regulatory framework are sufficiently distinctive that claims about other regulated domains would require separate empirical validation. Large-scale human evaluation panels are not employed as a primary evaluation instrument; human assessment is limited to spot-checking qualitative outputs for cases where automated metrics require validation, reflecting both the impracticality of human annotation at factorial experiment scale and the established reliability of RAGAS and G-Eval as automated evaluation tools.

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1.6 Significance of the Study

The significance of this study can be understood at four levels: its contribution to academic knowledge in NLP and information retrieval, its practical value for BFSI institutions deploying LLM-based systems, its relevance to regulators and policymakers overseeing AI adoption in the financial sector, and its broader contribution to public trust in AI-powered financial services.

At the academic level, the study makes three specific contributions to knowledge. First, it produces the first published factorial comparison of dense, sparse, and hybrid retrieval strategies on a BFSI-specific benchmark with hallucination rate and faithfulness as primary outcome measures. The significance of this contribution lies not merely in the results which will themselves be novel but in the experimental design: by simultaneously varying retrieval strategy, chunking strategy, retrieval depth, and citation grounding in a controlled factorial design, the study generates evidence about main effects and interaction effects that cannot be obtained from studies varying a single factor. This design rigour substantially increases the interpretive value of the findings. Second, the study extends the literature on chunking strategy in RAG by examining it as a co-varying factor alongside retrieval strategy in a domain-specific setting. Prior work has studied chunking in isolation (Smith and Troynikov, 2024; Hladěna et al., 2025) or has included it as one configuration among many without isolating its independent effect (Eibich et al., 2024). This study addresses that gap directly. Third, the study provides the first empirical measurement of prompt-level citation grounding as a controlled variable in BFSI QA, extending the citation grounding literature beyond the fine-tuning setting of (Ye et al., 2024) into the inference-only deployment context that characterises most production systems.

At the level of practitioner guidance, the study addresses a specific, pressing need. Financial institutions evaluating RAG architectures for compliance assistance, regulatory query resolution, and document QA applications currently lack domain-specific evidence to guide their retrieval strategy and pipeline configuration choices. The existing literature provides either general-domain results that may not transfer, or financial-domain results on isolated variables that do not support multi-dimensional trade-off analysis. This study's simultaneous four-way performance profile hallucination rate, faithfulness, latency, and token cost across all pipeline configurations directly enables the kind of evidenced deployment decision-making that practitioners require. The open-source release of the complete codebase and evaluation framework additionally lowers the barrier to independent replication and to adaptation of the pipeline for institution-specific document corpora, extending the study's practical impact beyond its own experimental results.

At the regulatory and policy level, the significance of this study lies in its contribution to the evidence base for AI governance in the financial sector. Regulators including the FCA, the European Securities and Markets Authority (ESMA), and the United States Office of the Comptroller of the Currency (OCC) have all published consultations or guidelines on the deployment of AI in regulated financial activities, consistently flagging hallucination risk and model explainability as primary concerns. A study that quantifies hallucination rates under different RAG configurations, identifies which configurations meet the reliability threshold necessary for use in compliance-sensitive workflows, and provides a reproducible methodology for institution-level verification of AI system performance provides precisely the kind of empirical evidence that regulatory frameworks need to move from principles to specific technical standards.

At the broadest level of public significance, the study matters because financial institutions that deploy more reliable AI systems provide better outcomes for the customers and communities they serve. Compliance failures triggered by AI-generated misinformation impose costs that ultimately manifest in regulatory fines, reputational damage, and reduced service quality. Advances in RAG reliability that allow financial institutions to deploy AI-powered tools that genuinely and consistently ground their outputs in authoritative documents rather than generating plausible-sounding confabulations contribute directly to financial services that are more trustworthy, more equitable, and more transparent. In this sense, the technical contribution of this study is in service of a broader social objective: ensuring that the deployment of AI in high-stakes regulated industries is grounded in rigorous evidence rather than commercial enthusiasm alone.

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1.7 Structure of the Study

The remainder of this thesis is organised into five further chapters, each of which advances the study from its conceptual foundation to its empirical findings and conclusions.

Chapter 2: Literature Review develops the theoretical and empirical context for the study in detail. Beginning with the broad landscape of LLM adoption in high-stakes professional domains, the chapter progressively narrows through hallucination taxonomy and its specific manifestations in financial settings, the architecture and evolutionary trajectory of RAG systems, the comparative properties of dense, sparse, and hybrid retrieval strategies and the indexing infrastructure that underlies them, the chunking strategy literature and its implications for retrieval-generation trade-offs, the role of citation grounding in generation-stage faithfulness, and the evaluation frameworks available for systematic RAG assessment. The chapter concludes with an explicit mapping of the research gaps that the study is positioned to address. The review draws on thirty-seven peer-reviewed and preprint sources and is structured to build a coherent argument from broad context to specific contribution, rather than to enumerate contributions in isolation.

Chapter 3: Methodology provides a detailed account of the experimental design, pipeline implementation, dataset preparation, and evaluation protocol. It describes the factorial structure of the experiment four independent variables, each with multiple levels, crossed over a fixed corpus and fixed model configuration and the statistical approach used to analyse the results. The modular architecture of the Python-based RAG pipeline is documented at implementation level, covering document ingestion and normalisation, each of the four chunking configurations, the construction of the dense TurboQuant index and the sparse BM25 inverted index and their hybrid RRF fusion, the retrieval depth parameterisation, the citation-grounding and standard prompt templates, and the integration of LLaMA 3.2-3B as the generation backbone. The evaluation framework is specified in full, including the five measurement dimensions, the tools used for each, and the procedure for automated evaluation using RAGAS and DeepEval.

Chapter 4: Results presents the empirical findings of the factorial experiment across all experimental conditions. The chapter is organised around the four research questions, reporting the measured values of each dependent variable hallucination rate, faithfulness, answer accuracy, response latency, and token cost for each retrieval strategy, chunking configuration, retrieval depth level, and citation-grounding condition, both individually and in combination. Results are supported by descriptive statistics, inferential tests, and visualisations that enable

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direct comparison across conditions. The chapter reports findings without interpretation, deferring explanatory discussion to Chapter 5.

Chapter 5: Discussion interprets the experimental findings in light of the theoretical framework established in the literature review and the specific research questions posed in Chapter 1. It examines where results are consistent with prior literature and where they diverge particularly in relation to the retrieval strategy performance rankings that prior general-domain studies might predict and offers explanations grounded in the specific characteristics of BFSI regulatory text and the FinanceBench task design. The chapter addresses each of the four research questions explicitly, discusses the practical implications of the findings for financial institution deployment decisions, identifies the limitations of the study and the sources of potential confound in the results, and proposes directions for future research that extend, replicate, or challenge the study's conclusions.

Chapter 6: Conclusion provides a concise synthesis of the study. It restates the research aim and objectives, summarises the principal findings, draws the key practical and theoretical conclusions that the evidence supports, and situates the contribution within the broader trajectory of RAG research in high-stakes professional domains. The chapter closes with a reflection on the implications for responsible AI deployment in the BFSI sector and the specific next steps that the research community and practitioner community might pursue on the basis of this study's findings.

Following Chapter 6, the thesis includes a complete References section listing all thirty-seven cited sources in full bibliographic detail, and an Appendices section containing the experimental configuration tables, prompt templates, evaluation output samples, and a guide to accessing the open-source codebase through its public repository.

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CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

The deployment of artificial intelligence in regulated industries is not a single problem it is a cluster of overlapping problems, each with its own theoretical grounding, its own empirical history, and its own set of open challenges. Before any experiment can be designed, any pipeline built, or any result interpreted, it is necessary to understand where the present study sits within this larger intellectual landscape: what has already been established, what remains contested, and what has simply not yet been asked. That is the purpose of this chapter.

This literature review is structured to build understanding progressively, from the broad context down to the precise technical questions that define the study's experimental design. It opens at the level of the forest the emergence of Large Language Models as a transformative technology in high-stakes professional domains, and the structural reliability problem that has accompanied that emergence before narrowing through successive layers to the specific trees: the mechanisms of hallucination in language models, the architecture and evolution of Retrieval-Augmented Generation as a grounding intervention, the comparative properties of dense, sparse, and hybrid retrieval strategies, the indexing infrastructure that makes large-scale retrieval tractable, the design of document chunking strategies, the role of citation grounding in constraining generation-stage confabulation, and the evaluation frameworks that make systematic measurement of these effects possible.

The scope of the review reflects a deliberate set of choices about what is and is not central to the research questions. This study is concerned specifically with the Banking, Financial Services, and Insurance (BFSI) sector, with regulatory policy question answering as the task type, and with inference-time pipeline design retrieval strategy, chunking, and prompting as the primary intervention. Model fine-tuning, graph-augmented retrieval, and multi-agent architectures are discussed where they illuminate the theoretical landscape or establish the scope boundary, but they are not treated as primary variables. This focus is not a limitation of ambition; it reflects the recognition that rigorous, controlled investigation of a well-defined problem contributes more durable knowledge than exploratory examination of a broader one.

The review draws on thirty-seven peer-reviewed and preprint sources. The primary literature was identified through systematic keyword searches across Google Scholar, arXiv, ACL Anthology, and the ACM Digital Library using the search terms "retrieval-augmented generation", "hallucination mitigation LLMs", "dense retrieval", "sparse retrieval BM25", "hybrid retrieval RRF", "chunking strategy RAG", "financial question answering NLP",

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"faithfulness evaluation RAG", "vector quantization indexing", and various combinations thereof. Results were screened first by title and abstract, with papers selected for full review if the abstract indicated direct relevance to at least one of the four research questions. Survey papers were prioritised for breadth; experimental papers were evaluated for methodological quality, dataset specificity, and recency. Blogs, white papers, and non-peer-reviewed industry reports were excluded. The resulting corpus spans from the foundational Transformer and BERT papers of 2017–2019 through to studies published in 2025 and 2026, reflecting both the foundational theory and the current state of the field.

The chapter is organised into nine sections beyond this introduction. Sections 2.2 through 2.4 establish the broad context: the rise of LLMs in high-stakes domains, the hallucination problem and its specific severity in BFSI settings, and the architecture and evolution of RAG as a mitigation framework. Sections 2.5 through 2.7 examine the specific retrieval strategies, indexing technologies, and graph-augmented approaches that constitute the experimental variables and their principal alternatives. Sections 2.8 and 2.9 address the pipeline design choices of chunking strategy and citation grounding. Section 2.10 reviews the evaluation frameworks through which all of the above are measured. Section 2.11 synthesises the literature into an explicit map of research gaps and situates the present study within them. A chapter-level summary is provided at the end of this document, drawing together the principal findings and preparing the transition to the methodology chapter.

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2.2 LLMs in Finance

The last decade has witnessed a fundamental transformation in how artificial intelligence systems process and generate natural language. The introduction of the Transformer architecture by (Vaswani et al., 2017) a model built entirely on self-attention mechanisms that dispensed with the recurrence and convolution of prior neural designs provided the computational foundation upon which an entirely new class of language systems would be built. What began as a breakthrough in machine translation quickly revealed itself as a general-purpose mechanism for language understanding, and within a few years, pre-trained language models such as BERT (Devlin et al., 2019) and the GPT family (Brown et al., 2020) demonstrated capabilities that reshaped expectations across research and industry alike. These models, collectively referred to as Large Language Models (LLMs), are trained on corpora of unprecedented scale encompassing hundreds of billions of tokens drawn from books, articles, code repositories, and the open web learning to encode rich statistical regularities of human language into billions of model parameters. The result is a class of system capable of answering questions, summarising documents, drafting correspondence, and engaging in multi-turn dialogue with a fluency that, to the casual observer, can be indistinguishable from human expertise.

These capabilities have not gone unnoticed in regulated industries. The Banking, Financial Services, and Insurance (BFSI) sector, long dependent on armies of analysts, compliance officers, and legal specialists to interpret vast bodies of regulatory text, has found in LLMs a compelling proposition: intelligent systems that can query policy documents, summarise regulatory obligations, identify compliance risks, and assist in customer-facing advisory roles, all at a fraction of the cost and latency of traditional approaches. Major global institutions have begun deploying LLM-based tools for tasks ranging from automated anti-money-laundering (AML) report drafting and capital adequacy analysis to insurance underwriting assistance and real-time regulatory query resolution. The market rationale is clear, regulatory compliance in a single large bank may involve thousands of pages of jurisdiction specific rules updated continuously by bodies such as the Basel Committee on Banking Supervision, the Financial Conduct Authority (FCA), the Federal Reserve, and the European Banking Authority (EBA). No human team can achieve consistent, instant recall across that corpus.

Yet within this promise lies a structural flaw that has proven obstinate. LLMs, regardless of scale, are generative systems trained to produce statistically plausible sequences of text. They do not retrieve facts; they reconstruct them from the probabilistic landscape encoded in their parameters during pre-training. When that parametric knowledge is outdated, incomplete, or simply absent as is frequently the case for highly specific regulatory provisions, recent legislative amendments, or institution specific policy documents the model does not flag its uncertainty. It generates a response that is internally coherent, grammatically polished, and often confidently framed, but factually incorrect. This phenomenon, widely termed hallucination, constitutes one of the most consequential unsolved problems in applied NLP (Tonmoy et al., 2024; Huang et al., 2025). In general consumer settings, a hallucinated recipe or a misattributed historical fact may be harmless or merely embarrassing. In BFSI environments, where every stated figure, regulatory reference, and policy condition carries potential legal and financial weight, such failures are genuinely dangerous. (Joshi, 2025) estimates that enterprise losses attributable to hallucination related incidents in BFSI contexts exceed two hundred and fifty million US dollars annually, while (Kang and Liu, 2023) provide empirical evidence that LLMs exhibit significantly elevated rates of factual error on financial reasoning tasks compared to general-domain benchmarks. A documented case involved a global bank's internal LLM powered chatbot generating a fictitious FATCA clause that was acted upon by compliance staff before the error was caught in audit; a second involved an LLM generated investment summary misstating the bond rating of a AAA-rated security.

The motivation for this study sits squarely within this context. The central argument is not that LLMs are unusable in BFSI environments their utility for language understanding and generation tasks is well-established but that deploying them in prompt-only configurations, without any grounding mechanism connecting model outputs to authoritative source documents, creates unacceptable reliability risk. The question this thesis investigates is therefore a practical engineering and empirical one: what pipeline configuration most effectively anchors LLM outputs to ground truth in BFSI policy question answering, and what are the measurable trade-offs between different design choices across the dimensions of hallucination, faithfulness, latency, and cost?

The sections that follow develop the conceptual scaffolding necessary to answer this question. They begin with a detailed examination of hallucination as a phenomenon, its causes, taxonomy and documented impact in the financial domain before moving through the architecture of Retrieval-Augmented Generation as a remediation approach, the specific retrieval strategies and indexing technologies that constitute the experimental variables of this study, the subtler pipeline design choices around chunking and citation grounding, the evaluation frameworks that make systematic measurement possible, and finally the research gaps that this study is positioned to address.

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2.3 Hallucinations in Large Language Models

Understanding hallucination requires first understanding the generative process that produces it. A modern LLM such as LLaMA (Touvron et al., 2023) is trained through a process of next-token prediction, given a sequence of input tokens, the model learns to assign probability distributions over possible continuations. This auto-regressive mechanism optimises for local token level plausibility rather than global factual coherence. The model has no internal fact checking module, no live connection to external knowledge bases, and no ability to distinguish between content it has encountered frequently in training and content it is effectively confabulating. (Brown et al., 2020) demonstrated through the GPT-3 experiments that as model scale increases, both capability and the propensity for fluent, confident confabulation increase together, scale suppresses some failure modes while amplifying others.

The most systematic treatment of hallucination types appears in (Tonmoy et al., 2024) , who surveyed more than thirty-two mitigation approaches across the literature and constructed a taxonomy spanning four principal failure categories. Factual contradictions arise when model output directly conflicts with verifiable reality stating for instance, that a particular regulation requires capital reserves of fifteen percent when the correct figure is eight percent. Contextual irrelevance refers to outputs that are factually benign in isolation but non-responsive to the specific question asked, often a symptom of retrieval failure in grounded systems. Logical inconsistency occurs when successive statements within a single response contradict one another, reflecting the absence of any coherent reasoning process behind surface fluency. Temporal disorientation arises when a model, whose parametric knowledge is frozen at a training cutoff, presents outdated information as current, a particularly acute risk in financial regulation, where rules are amended frequently and compliance obligations shift across legislative cycles.

Complementing this general taxonomy, (Huang et al., 2025) draw a critical distinction between intrinsic hallucination where model output contradicts the source material it was provided and extrinsic hallucination where the model introduces claims that can neither be confirmed nor refuted from the available evidence base. This distinction carries direct relevance for evaluation methodology: intrinsic hallucination is detectable by automated faithfulness metrics such as RAGAS (Es et al., 2024), while extrinsic hallucination requires either access to external ground truth or a trained judge model. Both types occur in BFSI policy QA contexts, but their relative frequency and consequences differ. Intrinsic hallucination in a RAG system where the model ignores or distorts retrieved passages represents a failure of the generation component; extrinsic

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hallucination indicates a failure of the retrieval component to surface relevant evidence, forcing the model to fall back on parametric inference.

The financial domain amplifies these risks in specific ways. (Kang and Liu, 2023) conducted one of the earliest systematic empirical investigations of LLM hallucination in financial reasoning tasks, finding that several prominent LLMs produced financially erroneous outputs at rates substantially higher than those observed on general domain benchmarks, particularly for questions involving numerical reasoning, regulatory cross-referencing, and temporal context. (Joshi, 2025) extends this analysis to BFSI operational settings, documenting a forty-seven million dollar legal settlement traceable to LLM generated misinformation in a financial advisory context, alongside a twenty two percent decline in customer satisfaction attributed to AI generated errors in a financial services chatbot deployment. The mechanisms by which financial text exacerbates hallucination risk are identifiable: regulatory documents are dense with domain specific terminology that is underrepresented in general pre-training corpora, numerical precision requirements are unforgiving in a way that prose summarisation tasks are not, and the consequence of a single misquoted figure or misidentified clause can trigger regulatory action rather than mere inconvenience.

Importantly, the failure mode is not purely one of model capability. (Barnett et al., 2024), in their analysis of seven systematic failure points in production RAG systems, identify a category they term "answer consolidation failure" where a model receives multiple retrieved passages that individually provide partial but incomplete answers, and must synthesise them into a coherent response. In regulatory QA, where a compliance question may require reconciling provisions from three different sections of a single regulation or across two separate legislative instruments, this failure mode is especially frequent. Their taxonomy of seven failure points which also includes insufficient context in retrieved passages, missing top-ranked relevant documents, extraction errors during document ingestion, incorrect format of extracted information, inability to handle unanswerable questions, and inadequate response consolidation provides a comprehensive engineering framework for understanding where in the pipeline failures occur, independent of the generation model's intrinsic limitations.

Taken together, this body of evidence establishes two conclusions that ground the present study. First, hallucination in LLMs is not a marginal edge case but a systematic structural property of auto-regressive generation, and its consequences in BFSI contexts are commercially and legally material. Second, mitigation strategies that operate solely at the model level prompt engineering, instruction tuning, or post-hoc correction have proven insufficient as complete solutions.

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2.4 RAG Architecture and Evolution

The foundational insight behind Retrieval-Augmented Generation is deceptively simple: rather than asking a language model to answer a question entirely from memory, supply it at inference time with the specific passages most likely to contain the correct answer, and instruct it to generate a response conditioned on that retrieved evidence. This approach decouples what the model needs to know from what it needs to be the model need not encode every regulatory provision in its parameters if the relevant provisions can be retrieved on demand from an authoritative document corpus. (Lewis et al., 2021) formalised this idea in the paper that introduced the RAG acronym to the NLP community, demonstrating that a retriever generator architecture on open-domain QA tasks substantially outperformed both retrieval only and generation only baselines, and establishing the retriever reader paradigm that has since become the dominant framework for knowledge intensive NLP.

The subsequent literature has substantially elaborated and diversified this original blueprint. (Gao et al., 2023) in the most widely cited survey of the RAG landscape, organise the evolution of the approach into three generations: Naive RAG, Advanced RAG, and Modular RAG. Naive RAG systems implement the original(Lewis et al., 2021) formulation directly: a fixed chunking strategy segments documents into passages, a single stage retriever selects the top-k passages given a query, and the generator produces a response conditioned on those passages concatenated into the prompt. This approach is simple, interpretable, and surprisingly competitive on benchmarks, but it is sensitive to retrieval quality in ways that compound at the generation stage if the retriever returns irrelevant or partially relevant passages, the generator has no mechanism to identify and filter them. (Gupta et al., 2024) extend this survey to document the broader RAG ecosystem across domains, noting that by 2023, RAG had become the default architectural pattern for enterprise LLM deployments requiring factual grounding.

Advanced RAG addresses Naive RAG's limitations through pre-retrieval, retrieval-stage, and post-retrieval enhancements. Pre-retrieval techniques include query expansion, where the original query is augmented with semantically related terms or reformulated as a hypothetical document (the Hypothetical Document Embedding approach, or HyDE, studied by (Eibich et al., 2024)); query decomposition, where complex multi-part questions are split into sub-queries addressed separately; and metadata enrichment, where document chunks are annotated with structured attributes such as section type, temporal validity, and regulatory scope to enable filtered retrieval. Retrieval-stage enhancements include re-ranking, where a cross-encoder model scores all retrieved candidates against the query with greater precision than the initial retriever afforded, and sentence window retrieval, where the matched passage is expanded to include surrounding sentences to provide generation-stage context. Post-retrieval enhancements include context compression, where a filtering model removes irrelevant segments from retrieved passages before they are passed to the generator, and iterative retrieval-generation loops, where the generator's initial output is used to formulate follow-up retrieval queries in a multi-hop process (Shao et al., 2023).

(Eibich et al., 2024) conducted one of the most careful controlled experiments within the Advanced RAG landscape, comparing twelve retrieval and post-retrieval configurations on open-domain corpora. Their results showed that HyDE and LLM-based re-ranking both yielded statistically significant improvements in retrieval precision, whereas Maximal Marginal Relevance (MMR) and commercial re-rankers offered negligible advantages over naive baselines in their experimental setting. Crucially, Sentence Window Retrieval emerged as the single highest-precision configuration across their benchmark. However, their corpus was deliberately general-domain, and the authors explicitly flag the question of whether these rankings hold for specialised regulatory text a question that this study is positioned to answer.

The third generation, Modular RAG, reconceptualises the pipeline as a set of interchangeable, composable modules that can be combined and sequenced in task-specific configurations. Rather than a fixed retrieve-then-generate sequence, Modular RAG systems can perform iterative retrieval, conditional retrieval (triggered only when the model's confidence in its parametric knowledge falls below a threshold), and hybrid pipelines that route different query types through different retrieval sub-systems. (Asai et al., n.d.) exemplify the Modular RAG philosophy in their Self-RAG framework, which trains a language model to emit special reflection tokens during generation indicating whether retrieval is needed, whether retrieved passages are relevant, and whether its own generated outputs are faithful to the retrieved evidence. Self-RAG demonstrates that adaptive retrieval selectively invoking the retriever rather than applying it unconditionally to every query can substantially improve both factuality and response coherence relative to fixed-retrieval baselines.

The anatomy of a RAG pipeline can be decomposed into five canonical stages, each of which introduces design choices with measurable downstream consequences. Document ingestion covers the loading, normalisation, and structuring of source documents for BFSI applications this primarily means PDF extraction from regulatory filings, annual reports, and policy manuals. Chunking segments those documents into retrieval units of appropriate granularity. Indexing converts chunks into a searchable representation, either as dense vector embeddings in a vector database or as sparse term weights in an inverted index. Retrieval uses a query representation to select the most relevant chunks from the index. Generation takes the selected chunks and the original query as input and produces a final answer. (Setty et al., 2024), in a study specifically focused on financial document QA, found through ablation that suboptimal retrieval rather than inadequate generation was the dominant source of inaccurate answers, with query expansion, metadata annotation, and embedding fine-tuning each independently improving retrieval precision in measurable ways. This finding directly motivates the experimental focus of the present study on the retrieval stage, specifically the comparison of dense, sparse, and hybrid retrieval strategies.

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2.5 Retrieval Strategies

The retrieval stage is the most consequential single component of a RAG pipeline for factual accuracy, and it is here that the three strategies under investigation in this study diverge most sharply in their theoretical properties, practical strengths, and domain-specific performance characteristics.

Sparse retrieval is the classical paradigm of information retrieval and has underpinned commercial search engines for decades. (Robertson and Zaragoza, 2009) document the mathematical foundations of BM25(Best Match 25) the ranking function that remains the dominant sparse retrieval algorithm in production systems. BM25 scores documents as a weighted function of term frequency normalised by document length and inverse document frequency across the corpus, capturing the intuition that terms which appear frequently in a document and rarely in the collection are strong indicators of relevance. The algorithm has no parameters that require corpus-specific training, operates purely on exact lexical match, and scales efficiently to very large document collections through inverted index structures. (Chen and Wiseman, 2023) demonstrate that BM25 can be made more powerful through learned query augmentation expanding the original query terms using a small end-to-end trained model improving recall on out-of-vocabulary terms. For BFSI regulatory text, sparse retrieval has a natural advantage: regulatory language is highly terminologically precise, with specific section references, clause numbers, defined terms, and numerical thresholds that dense semantic models may embed imprecisely. When a compliance officer asks about "Article 129(3) of the Capital Requirements Regulation," the exact retrieval of that article is a lexical matching task that BM25 handles by design.

Dense retrieval was established as a competitive paradigm for open-domain QA by (Karpukhin et al., 2020) in the Dense Passage Retrieval (DPR) paper, which demonstrated that a dual-encoder architecture where query and passage are independently encoded into a shared dense vector space using BERT (Devlin et al., 2019) could substantially outperform BM25 on natural question answering benchmarks, improving top-20 passage retrieval accuracy by nine to nineteen percentage points across several datasets. The representational advantage of dense retrieval lies in its ability to capture semantic similarity rather than exact lexical overlap: a query phrased as "what are the minimum reserve requirements for commercial banks" can retrieve a passage that uses the phrase "mandatory capital buffer thresholds for retail lending institutions" despite sharing no significant vocabulary. (Reimers and Gurevych, 2019) refined the efficient computation of sentence-level embeddings through Sentence-BERT, which uses siamese and triplet network structures to produce semantically meaningful fixed-length

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representations that can be compared using cosine similarity eliminating the need to pass query-document pairs jointly through the encoder at retrieval time and making large-scale approximate nearest-neighbour search tractable. Modern embedding models such as the qwen3-embedding:4b model (Zhang et al., 2025) used in this study extend this approach with instruction-tuned multilingual encoders trained on massive retrieval-specific corpora, achieving state-of-the-art performance on established embedding benchmarks.

The indexing infrastructure that enables dense retrieval at scale has historically been dominated by approximate nearest-neighbour libraries. The FAISS library (Douze et al., 2025), developed at Meta AI Research, provides a widely used collection of indexing algorithms including Flat indices for exact search, Inverted File (IVF) indices for approximate search, and Product Quantization (PQ) for memory-compressed retrieval. While FAISS has become the de facto standard for dense retrieval research, its product quantization approach introduces a trade-off between compression ratio and retrieval accuracy that is difficult to tune without corpus-specific calibration. A significant recent development that directly informs the implementation choices of this study is TurboQuant (Zandieh et al., 2026) accepted at ICLR 2026. TurboQuant introduces a two-stage, data-oblivious vector quantization algorithm that achieves near-optimal distortion rates within a constant factor of the information-theoretic lower bound established by Shannon's source coding theory. The first stage termed PolarQuant randomly rotates input vectors using a Haar-distributed orthogonal matrix, inducing a concentrated Beta distribution on coordinates that makes the distribution predictable regardless of the original input. Optimal scalar quantizers are then applied per coordinate. The second stage applies a 1-bit Quantized Johnson-Lindenstrauss (QJL) transformation to the quantization residual, producing an unbiased inner product estimator that corrects the bias introduced by the MSE-optimal first stage. In nearest-neighbour search experiments, TurboQuant outperforms FAISS-based product quantization on recall@k benchmarks while reducing indexing time to virtually zero the method requires no offline calibration, no training data, and no codebook construction. This positions TurboQuant as a particularly attractive indexing backend for the modular, reproducible RAG pipeline of this study, and it is adopted as the dense indexing layer in place of FAISS.

The complementarity between sparse and dense retrieval has motivated extensive investigation of hybrid approaches that combine the precision of BM25 on exact keyword queries with the semantic coverage of dense retrieval. Reciprocal Rank Fusion (RRF), introduced by (Cormack et al., 2009), provides an elegant, parameter-free method for merging ranked lists from multiple retrieval systems. For each document, RRF computes a score equal to the sum of reciprocal

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ranks across all constituent retrievers, scaled by a constant k that moderates the influence of lower-ranked documents. This rank-based fusion avoids the score normalisation problem that arises when trying to linearly combine scores from systems with incompatible scaling BM25 scores and cosine similarity values inhabit different ranges and consistently outperforms individual retrievers as well as the classical Condorcet fusion method in (Cormack et al., 2009) original TREC experiments.

Subsequent work has confirmed and extended these findings in RAG-specific settings. (Mala et al., 2025), in a comparative analysis explicitly targeting hallucination reduction, find that hybrid retrieval combining dense and sparse signals via RRF achieves a mean average precision at rank three (MAP@3) of 0.897 against BM25 and dense baselines both below 0.75. (Akarsu et al., 2026), in the most recent benchmarking study of retrieval strategies specifically on financial text-and-table documents, find that two-stage hybrid retrieval with neural re-ranking achieves Recall@5 of 0.816 a seventeen percentage point improvement over un-reranked hybrid retrieval and importantly observe that BM25 alone outperforms state-of-the-art dense retrieval on their financial corpus, a finding that challenges the assumption that dense methods universally dominate in specialised domains. Similarly, (Kim et al., 2025) find that hybrid RRF fusion improves recall over individual methods by fifteen to thirty percent on financial QA corpora. These findings collectively motivate the inclusion of all three retrieval strategies as experimental conditions in this study and underscore the importance of empirical comparison rather than assumption-driven selection.

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2.6 Graph-Based and Advanced RAG

While the flat-retrieval paradigm where documents are segmented into independent chunks and retrieved without regard to the relational structure between them is the dominant baseline in RAG research, an alternative lineage of work represents document knowledge as structured graphs and performs retrieval by traversing graph substructures. This approach is particularly motivated by the nature of financial regulatory text, where compliance obligations frequently arise from the interaction of provisions across multiple sections, instruments, or jurisdictions rather than from any single self-contained passage.

(Barry et al., 2025) propose two graph-augmented retrieval strategies FactRAG and HybridRAG and evaluate them directly on the FinanceBench benchmark (Islam et al., 2023). In their system, document content is represented as a knowledge graph where nodes correspond to regulatory concepts, entities, and provisions, and edges encode relationships such as dependency, exception, and cross-reference. Retrieval proceeds by identifying relevant graph substructures rather than flat text chunks, allowing the system to surface the interconnected set of provisions necessary to answer complex compliance questions. Their results show a six percent reduction in hallucination rate and an eighty percent reduction in token consumption relative to standard flat RAG, with cross-document comparison complexity reduced from $O(n^2)$ to $O(k \cdot n)$ for regulatory cross-referencing tasks such as comparing DORA guidelines against FFIEC frameworks.

These results are suggestive of the potential for graph-structured retrieval in financial settings. However, several methodological limitations complicate their interpretation. The graph construction process introduces substantial pre-processing overhead that is not quantified relative to flat-retrieval baselines, making it difficult to assess whether the hallucination reduction derives from graph structure per se or from the additional engineering effort in document preparation. More critically, (Barry et al., 2025) do not compare their system against dense or sparse flat-retrieval baselines under controlled conditions, leaving open the question of how much of the performance improvement is attributable to graph structure versus other pipeline differences such as re-ranking, passage expansion, or prompt engineering. The present study explicitly excludes graph-augmented retrieval from its experimental comparison for this reason the additional structural complexity constitutes a confounding factor that is better addressed in a dedicated study but acknowledges it as an important architectural direction for future investigation.

The broader Advanced RAG literature provides additional context for the scope choices of this study. (Eibich et al., 2024) find that HyDE which generates a hypothetical answer to the query

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and uses the embedding of that hypothetical as the retrieval query delivers consistent precision gains on open-domain benchmarks, but raises concerns about latency and cost for high-throughput production deployments. Self-RAG (Asai et al., n.d.) demonstrates that adaptive retrieval, where the model decides per-query whether retrieval is beneficial, can outperform unconditional retrieval-based baselines and unconditional no-retrieval baselines simultaneously but requires fine-tuning of the language model backbone, introducing dependencies on training infrastructure that are incompatible with the inference-only scope of this study. (Shao et al., 2023) show that iterative retrieval, where successive rounds of generation inform follow-up retrieval queries, is particularly beneficial for multi-hop reasoning tasks but at the cost of latency that may be prohibitive in real-time compliance assistance deployments.

These design choices collectively define the experimental frontier against which this study situates itself: a systematic empirical investigation of the most practically deployable configurations flat dense, sparse, and hybrid retrieval with controlled chunking and prompting variables evaluated against a BFSI-specific benchmark, producing results that are directly actionable by financial institutions without requiring model fine-tuning or graph construction infrastructure.

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2.7 The FinanceBench Dataset

A significant methodological challenge facing any study of RAG performance in the financial domain is the absence of suitable evaluation benchmarks. General-domain open QA benchmarks such as Natural Questions, TriviaQA, or HotpotQA were constructed from Wikipedia and web sources, and the questions they contain drawn from encyclopaedic knowledge do not reflect the structural and terminological characteristics of regulatory policy text. Evaluation on these benchmarks, while useful for demonstrating general capability, provides no reliable evidence of performance in BFSI-specific deployments.

(Islam et al., 2023) address this gap with FinanceBench, a purpose-built financial question answering dataset constructed from real corporate filings, including SEC 10-K annual reports, 10-Q quarterly reports, and earnings call transcripts from publicly traded companies. FinanceBench provides annotated question-answer pairs drawn from these documents, with questions targeting both factual extraction tasks (specific numerical values, named entities, and regulatory references) and reasoning tasks (calculations derived from multiple reported figures, year-over-year comparisons, and trend identification). The grounded nature of the benchmark all answers are explicitly traceable to specific passages in the source documents makes it particularly suitable for evaluating RAG faithfulness: a model that produces the correct answer but cannot attribute it to an identifiable retrieved passage is treated as hallucinating, even if the answer happens to be correct.

FinanceBench has rapidly become the standard evaluation corpus for BFSI-focused RAG research. (Barry et al., 2025) use it to evaluate their graph-augmented retrieval system; (Setty et al., 2024) use it in their retrieval optimisation study; and (Akarsu et al., 2026) use a closely related financial text-and-table corpus with comparable question types. The dataset's combination of authentic document provenance, annotated grounding, and coverage of numerical reasoning tasks makes it the most appropriate primary benchmark for the experimental evaluation in this study.

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2.8 The Effect of Chunking Strategy

Among the pipeline design choices that determine RAG system performance, document chunking occupies a curious position in the research literature: it is widely acknowledged as consequential but rarely studied as a primary experimental variable with the same rigour as retrieval strategy or model selection. Chunking determines the granularity of the units the retrieval system operates on, and this granularity mediates a fundamental trade-off between retrieval precision and generation-stage context.

When chunks are small say, a single sentence or a fixed window of 128 tokens, each chunk is relatively semantically specific, and the embeddings of nearby chunks are more discriminably different from one another, enabling precise retrieval of the exact passage relevant to a query. However, small chunks frequently lack the surrounding context necessary for the generation model to produce a coherent, complete answer: a retrieved sentence stating "the minimum ratio is eight percent" provides no information about which ratio, which entity type it applies to, under which regulatory framework, or in which jurisdictional context. When chunks are large covering several paragraphs or entire document sections they provide rich generation context but dilute embeddings by mixing multiple distinct topics, reducing the probability that the retrieval system will correctly rank the chunk above less relevant alternatives.

(Smith and Troynikov, 2024) formalise this trade-off in an empirical study of chunking strategies across multiple retrieval benchmarks. Their central finding is that semantic chunking where boundaries are placed at detected topic transitions rather than at fixed token counts consistently outperforms fixed-size chunking on retrieval precision. The intuition is straightforward: fixed-size chunking cuts arbitrarily across semantic boundaries, producing chunks that partially cover multiple topics and embed as a muddled average, whereas semantic chunking produces chunks that are each topically coherent and therefore produce more discriminating embeddings.

(Hladěna et al., 2025) extend this investigation to a broader range of domains and embedding models, reaching the important negative finding that no single chunk size is universally optimal: effectiveness depends jointly on the domain, the query type, and the specific embedding model. For regulatory text which tends to be structured into numbered articles and sub-clauses with relatively clear topical boundaries semantic chunking provides a natural alignment between chunk boundaries and regulatory unit boundaries. For earnings call transcripts which are discursive and less structured sentence-window chunking may perform differently. This domain dependence makes comparative experimentation across chunking

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strategies, with domain-specific corpora, methodologically necessary rather than merely interesting.

The sentence-window retrieval approach, studied by (Eibich et al., 2024) and implemented in this study as one of four chunking conditions, represents a hybrid strategy that partially decouples retrieval precision from generation context. Retrieval operates on small, precise chunks (typically individual sentences), but when a chunk is selected, the system expands the retrieved context to include a window of surrounding sentences before passing it to the generator. This allows the retrieval system to maintain the discriminative advantage of small chunks while ensuring the generator receives enough context to produce coherent, grounded responses. (Eibich et al., 2024) report that Sentence Window Retrieval achieves the highest precision of all configurations tested in their ARAGOG study, making it a strong candidate for financial regulatory text where precise clause-level retrieval and adequate context provision are simultaneously required.

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2.9 Citation Grounding

A dimension of RAG hallucination mitigation that has received growing attention in the recent literature, but remains understudied as a controlled experimental variable, is citation grounding: the explicit requirement that every factual claim in a generated response be attributed to a specific retrieved passage. This requirement operates on the generation stage rather than the retrieval stage, and its theoretical motivation is distinct from retrieval quality improvements. Even when high-quality relevant passages are retrieved and injected into the prompt, the language model may incorporate information from its parametric memory information not present in any retrieved passage alongside retrieved evidence, particularly for queries that touch areas of its pre-training data. A citation grounding constraint forces the model to restrict itself to claims it can explicitly anchor in the retrieved material.

(Maynez et al., 2020) provide the conceptual foundation for this distinction in their landmark study of faithfulness and factuality in abstractive summarisation. Analysing outputs from multiple summarisation models on the CNN/DailyMail benchmark, they find that intrinsic hallucination where model output contradicts the source document occurs at rates of ten to thirty percent depending on model architecture, and that longer, higher-fluency outputs are often associated with lower faithfulness rather than higher quality. They propose the faithfulness metric measuring what fraction of generated claims are directly attributable to source passages as a more appropriate evaluation criterion than surface-level quality measures for grounded generation tasks. This metric is directly operationalised in the RAGAS framework (Es et al., 2024) used as a primary evaluation tool in this study.

(Ye et al., 2024) address the citation generation problem through AGREE (Adaptation for GRounding EnhancEment), a learning-based framework that fine-tunes LLMs to self-ground their responses by generating accurate passage-level citations alongside their answers. Rather than relying on zero-shot or few-shot prompting to elicit citations, AGREE constructs training data automatically from unlabelled queries using a Natural Language Inference (NLI) model to identify which retrieved passages entail each claim in the model's output, using this attribution signal to supervise the model in generating explicit sentence-level citations. Evaluated against prompting-based and post-hoc citation baselines, AGREE achieves improvements of more than twenty percent in both citation recall and citation precision, and demonstrates strong generalisation to out-of-distribution domains including enterprise customer support corpora.

In the context of this study, citation grounding is examined not through model fine-tuning which would require departing from the inference-only pipeline design but through prompt-

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level instruction: a system prompt that explicitly instructs the model to tag each factual claim with the passage identifier from which it originates, and to generate no claim that cannot be so tagged. This approach is simpler than AGREE and does not require any training, but it is practically important to measure its effect on faithfulness scores relative to a standard RAG prompt without citation instructions. The research questions this addresses RQ4 is whether the marginal overhead of a citation grounding instruction produces a measurable faithfulness improvement on the FinanceBench evaluation set, and whether that improvement is consistent across different retrieval strategy and chunking configurations.

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2.10 Evaluation Frameworks for RAG Systems

The evaluation of RAG systems is a non-trivial problem that has generated a dedicated sub-literature. The core difficulty is that RAG evaluation requires simultaneously assessing multiple components retrieval quality, faithfulness of the generated answer to retrieved content, factual accuracy against external ground truth, and system-level efficiency that are measured by different tools and reflect different potential failure modes. A system that retrieves perfectly but generates unfaithfully, a system that retrieves poorly but happens to generate correctly from parametric memory, and a system that performs reliably but at prohibitive latency cost, all require different diagnostic frameworks.

(Es et al., 2024) address this challenge through the RAGAS framework, an automated evaluation library that computes four complementary metrics from the components of a RAG response. Context precision measures what fraction of the retrieved passages are relevant to the query, providing a signal of retrieval quality. Context recall measures what fraction of the ground-truth answer's claims are recoverable from the retrieved passages, capturing retrieval completeness. Faithfulness measures what fraction of the model's generated claims are directly attributable to retrieved passages, operationalising the (Maynez et al., 2020) definition. Answer relevance measures the semantic alignment between the generated response and the original query. By decomposing evaluation across these dimensions, RAGAS enables precise diagnosis of where pipeline performance degrades whether at retrieval, at generation, or at the interface between them. The library is implemented as a Python package and supports automated evaluation using a judge LLM, making it practical for large-scale factorial experiments.

(Liu et al., 2023) introduce G-Eval as a complementary evaluation approach, using DeepSeek-R1-8B as an automated judge to assess natural language generation quality across dimensions including coherence, consistency, fluency, and relevance. G-Eval is specifically designed to align with human judgements more closely than classical surface metrics such as BLEU or ROUGE, addressing the well-documented problem that lexical overlap metrics are poor proxies for factual accuracy. In the context of this study, G-Eval is implemented through the DeepEval framework (AI, 2023), which wraps G-Eval alongside hallucination rate measurement into a practical evaluation pipeline compatible with the modular RAG architecture.

(Sivasothy et al., 2024) identify critical limitations of existing RAG evaluation frameworks through RAGProbe, a stress-testing tool that generates question variations paraphrases, negations, multi-hop reformulations to probe the robustness of RAG pipelines beyond single-question evaluations. Their results are sobering: across five open-source RAG systems,

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RAGProbe identifies a ninety-one percent failure rate on multi-document spanning questions and a seventy-eight percent failure rate on questions combining information from multiple sections of a single document. Both question types are commonplace in BFSI regulatory QA, where a compliance question may require reconciling provisions from across multiple sections of a regulation or comparing requirements across multiple regulatory instruments. These failure rates are substantially higher than those reported on standard single-passage retrieval benchmarks, suggesting that current evaluation methodologies significantly overestimate real-world RAG reliability. The multi-document and multi-section question types are explicitly represented in the FinanceBench benchmark used in this study, ensuring that evaluation is conducted on questions that are representative of the actual BFSI compliance QA challenge.

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2.11 Research Gaps and Positioning of the Study

The literature reviewed in the preceding sections reveals a field that is productive and rapidly advancing but that contains several significant gaps, particularly in the specific intersection of BFSI policy question answering, RAG pipeline design, and systematic empirical evaluation.

The first and most significant gap is the absence of a controlled, side-by-side empirical comparison of dense, sparse, and hybrid retrieval strategies evaluated on a BFSI-specific benchmark with hallucination rate and faithfulness score as primary outcome measures. The majority of RAG retrieval comparisons in the literature are conducted on general-domain corpora (e.g., (Eibich et al., 2024); (Mala et al., 2025)), and those that do focus on financial text either do not include all three retrieval strategy types (Setty et al., 2024) or evaluate a different set of pipeline variables (Barry et al., 2025). The provocative finding of (Akarsu et al., 2026) that BM25 outperforms dense retrieval on financial documents has not been replicated or examined under the controlled conditions this study will apply.

The second gap concerns chunking strategy. Despite the well-established theoretical importance of chunk granularity for retrieval precision and generation coherence, no published study treats chunking strategy as a primary experimental variable alongside retrieval strategy in a factorial design. (Smith and Troynikov, 2024) and (Hladěna et al., 2025) study chunking in isolation, and (Eibich et al., 2024) include sentence-window retrieval as one configuration among many, but none simultaneously vary chunking and retrieval strategy across a BFSI-specific benchmark. The interaction effects between these two variables whether, for instance, semantic chunking confers the same advantage over fixed-size chunking for sparse retrieval as it does for dense retrieval remain yet to be investigated.

The third gap concerns citation grounding. While (Ye et al., 2024) demonstrate through AGREE that fine-tuned citation grounding substantially improves faithfulness, the effect of prompt-level citation grounding instructions without any model fine-tuning has not been isolated as a controlled variable in a factorial experiment. Whether a simple prompt instruction to cite retrieved passages produces a measurable faithfulness improvement, and whether this effect is consistent across retrieval strategies, is an empirically open question.

The fourth gap is the absence of a simultaneous four-way performance profile encompassing hallucination rate, faithfulness, response latency, and per-query token cost across a matrix of pipeline configurations. Existing studies typically report one or two of these metrics, making it impossible for practitioners to make deployment decisions that explicitly account for the latency-accuracy and cost-accuracy trade-offs that are material in production financial systems.

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This study is designed to address all four gaps through a unified factorial experimental design applied to the FinanceBench corpus. By holding the generation model (LLaMA 3.2-3B) and embedding model (qwen3-embedding:4b) constant across all conditions and systematically varying retrieval strategy, chunking approach, retrieval depth, and citation grounding, the study produces results that isolate the causal contribution of each design choice. The adoption of TurboQuant as the dense vector indexing layer additionally enables a comparison of indexing efficiency under realistic hardware constraints, extending the practical scope of the evaluation beyond accuracy metrics alone. The result will be, to the knowledge of the author, the first published factorial comparison of these pipeline variables on a BFSI benchmark, and the first study to provide a simultaneous four-way performance profile that practitioners can use to make evidenced deployment decisions.

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2.12 Chapter Summary

This chapter has reviewed the literature pertinent to the design, evaluation, and contextualisation of Retrieval-Augmented Generation pipelines for hallucination mitigation in BFSI policy question answering. Across nine substantive sections, drawing on thirty-seven sources spanning foundational AI research, information retrieval, NLP evaluation methodology, and financial domain studies, a coherent picture has emerged of a field at a critical juncture technically mature enough to produce production-scale RAG deployments, but not yet systematic enough to provide evidenced guidance on the design choices that most consequentially determine those deployments' reliability.

The opening sections established that Large Language Models, built on the Transformer architecture introduced by (Vaswani et al., 2017) and scaled through successive generations from BERT (Devlin et al., 2019) to the GPT family (Brown et al., 2020) and the open-weight LLaMA series (Touvron et al., 2023), have achieved genuine and commercially significant capabilities in language understanding and generation. The BFSI sector has been an early and aggressive adopter of these capabilities, motivated by the scale and complexity of regulatory documentation that no human team can monitor comprehensively. However, the literature is unambiguous that deploying LLMs in prompt-only configurations for high-stakes financial tasks introduces hallucination risk at rates and consequences that are commercially, legally, and reputationally unacceptable. (Kang and Liu, 2023) established the financial domain-specific elevation of hallucination rates; (Joshi, 2025) quantified their operational consequences; (Barnett et al., 2024) provided a structured engineering taxonomy of the failure modes through which they arise. The combined weight of this evidence establishes that hallucination mitigation is not an optional enhancement for BFSI LLM deployments but a precondition for responsible adoption.

The review of Retrieval-Augmented Generation, building on (Lewis et al., 2021) foundational RAG paper and the comprehensive survey of (Gao et al., 2023), traced the evolution of the approach from Naive RAG through Advanced RAG to Modular RAG, establishing the key insight that retrieval quality is the dominant determinant of end-to-end system faithfulness more influential than generation model capacity or prompt design under standard conditions (Setty et al., 2024). This finding directly motivates the experimental priority of the present study: if retrieval quality matters most, then systematic empirical comparison of retrieval strategies under controlled conditions is the most leveraged research contribution the study can make.

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The examination of retrieval strategies revealed a theoretically grounded landscape of complementary methods with distinct strengths. Sparse retrieval via BM25 (Robertson and Zaragoza, 2009) offers lexical precision and term-exact recall that is particularly well-suited to regulatory text with precise terminological conventions. Dense retrieval, established by (Karpukhin et al., 2020) through DPR and refined through successive embedding models including Sentence-BERT (Reimers and Gurevych, 2019) and the modern qwen3-embedding:4b (Zhang et al., 2025), captures semantic similarity across paraphrastic variation in query and document phrasing. Hybrid retrieval via Reciprocal Rank Fusion (Cormack et al., 2009) combines both modalities without requiring score normalisation, and multiple recent studies ((Mala et al., 2025); (Barnett et al., 2024); (Kim et al., 2025)) have shown consistent accuracy gains from fusion over either individual method. The surprising finding of (Akarsu et al., 2026) that BM25 alone outperforms dense retrieval on financial documents serves as an important caution against the default assumption that semantic density is universally superior to lexical precision, and it reinforces the necessity of domain-specific empirical evaluation rather than technique extrapolation from general-domain benchmarks.

The discussion of indexing infrastructure highlighted TurboQuant (Zandieh et al., 2026) as a theoretically significant advancement over the FAISS product quantization baseline. Its data-oblivious design, near-optimal distortion rate, and near-zero indexing time collectively make it a compelling engineering choice for the modular RAG pipeline of this study, and it replaces FAISS as the dense indexing layer in the experimental implementation. This substitution is not merely a technical implementation choice; it represents an alignment of the study with the current frontier of indexing technology and enables a direct comparison of indexing efficiency as a secondary outcome measure.

The literature on chunking strategy reviewed through the work of (Smith and Troynikov, 2024), (Hladěna et al., 2025), and (Eibich et al., 2024) confirms the theoretical expectation that chunk granularity and boundary placement materially affect retrieval precision and generation context quality. Semantic chunking and sentence-window retrieval emerge as strong candidates for regulatory text, but no prior study has treated chunking as a primary variable in a factorial design alongside retrieval strategy, leaving interaction effects between these two design choices unmeasured.

Citation grounding, examined through the foundational faithfulness framework of (Maynez et al., 2020) and the AGREE system of (Ye et al., 2024), emerged as a generation-stage intervention that has demonstrated substantial faithfulness improvements when implemented through fine-tuning. Whether equivalent improvements are achievable through prompt-level

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instruction alone without any model training is an empirically open question that this study addresses through a controlled citation grounding condition. The RAGAS evaluation framework (Es et al., 2024) and the DeepEval-mediated G-Eval approach (AI, 2023; Liu et al., 2023) provide the evaluation instrumentation through which this and all other experimental conditions are measured. The RAGProbe stress-testing findings of (Sivasothy et al., 2024) have additionally informed the selection of evaluation questions, ensuring representation of the multi-section and multi-document question types that expose the failure modes most relevant to compliance QA.

Finally, the structured analysis of research gaps in Section 2.11 established four specific areas where the present literature provides incomplete guidance: the absence of a controlled comparison of all three retrieval strategies on a BFSI benchmark with hallucination rate and faithfulness as primary outcomes; the absence of a factorial design treating chunking strategy and retrieval strategy as co-varying experimental factors; the yet to be investigated effect of prompt-level citation grounding instructions on faithfulness; and the absence of a simultaneous four-way performance profile capturing accuracy, faithfulness, latency, and cost across pipeline configurations. The study reported in the subsequent chapters is designed to address all four gaps.

Taken together, the literature reviewed in this chapter supports three conclusions that have guided every aspect of the study design. First, RAG is the correct architectural framework for hallucination mitigation in BFSI policy QA, it is the most mature, most empirically validated, and most practically deployable solution available. Second, within RAG, retrieval strategy is the single design choice most consequentially associated with output quality, and it is the design choice that has received the least systematic domain-specific evaluation in the financial context. Third, BFSI is not a setting where general-domain benchmarks and general-domain experimental results can be safely extrapolated, the domain's terminological precision, numerical rigour, and document structure make it sufficiently distinctive that empirical investigation on domain-representative data is a methodological necessity rather than merely desirable.

The methodology chapter that follows operationalises these conclusions into a concrete experimental design, defining the corpus, the pipeline configurations, the evaluation protocol, and the statistical analysis plan through which the four research questions identified in Chapter 1 will be answered.

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CHAPTER 3
RESEARCH METHODOLOGY

3.1 Introduction to the Chapter

This chapter describes, in precise and replicable detail, the methods through which this study addresses its five stated objectives and answers its four research questions. Research methodology is not merely a procedural formality; it is the intellectual contract between the researcher and the reader it specifies exactly what was done, how it was done, why it was done that way, and how the design choices ensure that the conclusions drawn are attributable to the manipulated variables rather than to hidden confounders. This chapter honours that contract.

The chapter is structured to follow the natural sequence of the research process. It begins with the philosophical grounding of the research design the epistemological position that justifies a quantitative, controlled experimental approach for this particular research problem. It then introduces the dataset and its characteristics, before walking step-by-step through each stage of the RAG pipeline implementation: document ingestion and preprocessing, chunking, indexing (dense, sparse, and hybrid), retrieval, prompt construction, and generation. The experimental design is then specified through a formal factorial variable matrix, followed by a complete description of the evaluation framework and the statistical analysis plan. The chapter concludes by mapping each research objective to the specific methodological stage that addresses it, and by acknowledging the principal ethical considerations and limitations of the design.

Where the methodology involves established techniques, references are provided and the reasons for selecting a particular approach over its alternatives are explained. No design choice is presented as the only option; each is justified. Readers who wish to replicate this study on a different corpus, with a different language model, or with additional pipeline components should find this chapter sufficient to do so.

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3.2 Research Design and Philosophical Underpinning

This study adopts a quantitative, controlled experimental design grounded in a post-positivist epistemological position. Post-positivism holds that objective reality exists and is knowable, but that our observations of it are always theory-laden and imperfect, and therefore that knowledge is established through the convergence of multiple measurements under controlled conditions rather than through any single definitive test. This is an appropriate philosophical position for empirical NLP systems evaluation: the concepts being measured hallucination rate, faithfulness, retrieval precision are real and meaningful, but any single measurement is subject to noise from query phrasing, model stochasticity, and evaluation instrument limitations.

The quantitative experimental approach is justified over qualitative or mixed-methods alternatives by the nature of the research questions. Each of the four RQs asks which of two or more conditions produces a superior outcome on a numerically measurable dimension. RQ1 asks which retrieval strategy produces lower hallucination rates and higher faithfulness scores. RQ2 asks how chunking strategy affects retrieval precision numerically. RQ3 asks at what value of k diminishing returns appear. RQ4 asks whether citation grounding produces a statistically significant faithfulness improvement. None of these questions is answerable through interpretive analysis, case study, or narrative synthesis; all require controlled numerical measurement and inferential statistical testing.

A factorial experimental design is chosen over a one-factor-at-a-time approach because it enables the detection of interaction effects on cases where the impact of one variable depends on the level of another. For example, the advantage of semantic chunking over fixed-size chunking may be larger for dense retrieval (which depends on embedding quality and is therefore more sensitive to chunk coherence) than for sparse BM25 retrieval (which operates on lexical features that are less influenced by semantic boundary placement). A one-factor-at-a-time design that varied chunking while holding retrieval strategy constant at an arbitrary level would miss this interaction and could lead to recommendations that are only valid in a specific, unstated context. The factorial design makes all interactions detectable and all main effects estimable while controlling for confounding.

All other pipeline components the LLM backbone, the embedding model, the prompt template structure, the evaluation tools, and the hardware environment are held constant across all experimental conditions. This ensures that observed differences in dependent variable values are attributable to the manipulated independent variables rather than to incidental variation in components that are not of primary interest.

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3.3 Dataset Description: FinanceBench

The primary evaluation corpus for this study is FinanceBench (Islam et al., 2023), a purpose-built financial question answering benchmark constructed from real corporate disclosure documents filed with the United States Securities and Exchange Commission (SEC). FinanceBench was selected as the evaluation corpus for four reasons that together make it uniquely suitable for this study.

First, the documents are authentic rather than synthetic. The corpus consists of 10-K annual reports, 10-Q quarterly reports, and earnings call transcripts from publicly traded companies across diverse sectors including banking, insurance, asset management, and financial technology. These documents exhibit the structural and terminological characteristics of real-world BFSI regulatory and disclosure text: dense numerical tables, defined terms with precise regulatory meaning, cross-referenced sections, and a combination of narrative prose and financial statements. Evaluation on this corpus generates findings that are ecologically valid for real BFSI deployment scenarios.

Second, all question-answer pairs are explicitly grounded. Each question in the dataset is paired with a ground-truth answer and, critically, with the source passage from which that answer is derivable. This grounded structure enables evaluation of both retrieval quality (whether the relevant passage was retrieved) and generation faithfulness (whether the model's answer is attributable to a retrieved passage), which is essential for measuring the specific hallucination mitigation effects this study investigates.

Third, the dataset includes question types that stress-test RAG pipelines in ways that general-domain benchmarks do not. Questions require factual extraction of specific numerical values, calculations derived from multiple reported figures (e.g., computing year-over-year revenue growth from two separate table entries), temporal reasoning across multiple reporting periods, and comparison of figures across different sections of a single filing. These question types are representative of real compliance and financial analysis queries and are precisely the types that (Sivasothy et al., 2024) identify as causing the highest failure rates in RAG systems.

Fourth, FinanceBench is publicly available without licensing restrictions, ensuring that the study's results are fully reproducible by other researchers using the same corpus.

The dataset comprises 150 annotated question-answer pairs drawn from 25 company filings. For this study, all 150 questions are used as the evaluation set. Each of the 25 source documents is processed through the full pipeline separately, with chunking and indexing applied per-document, and retrieval operating over each document's chunk index for the questions corresponding to that filing. This document-scoped retrieval design reflects the realistic BFSI

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deployment pattern in which a compliance officer queries a specific regulatory instrument or filing rather than a merged cross-document index.

Table 3.1 below summarises the key characteristics of the FinanceBench corpus used in this study.

Table 3.1: FinanceBench Corpus Characteristics

Attribute	Value
Total question-answer pairs	150
Source documents	25 SEC filings (10-K, 10-Q, earnings transcripts)
Document types	Annual reports (10-K), Quarterly reports (10-Q), Earnings call transcripts
Average document length	~45,000 tokens
Question types	Factual extraction, numerical calculation, temporal reasoning, cross-section comparison
Ground truth	Answer text + source passage identifier for each QA pair
Licence	Publicly available, MIT licence
Primary use in this study	Evaluation corpus; retrieval, generation, and automated metric computation

3.4 Overview of the Experimental Workflow

Before describing each pipeline stage in detail, it is useful to view the complete workflow at a high level. The experimental pipeline comprises five sequential functional stages, each of which transforms inputs into outputs that feed the next stage. These stages are: (1) Document Ingestion and Preprocessing, (2) Chunking, (3) Indexing, (4) Retrieval, and (5) Prompt Construction and Generation. The evaluation framework operates post-generation, comparing generated outputs against ground-truth answers and retrieved passages using automated metrics.

The pipeline is implemented as a modular Python architecture in which each stage is an interchangeable component. The three retrieval strategies dense, sparse, and hybrid are implemented as separate retrieval modules that share identical upstream (ingestion, chunking) and downstream (prompt construction, generation, evaluation) components. This modularity ensures that performance differences observed across retrieval strategies cannot be attributed differences in any other pipeline component as shown in figure 3.1

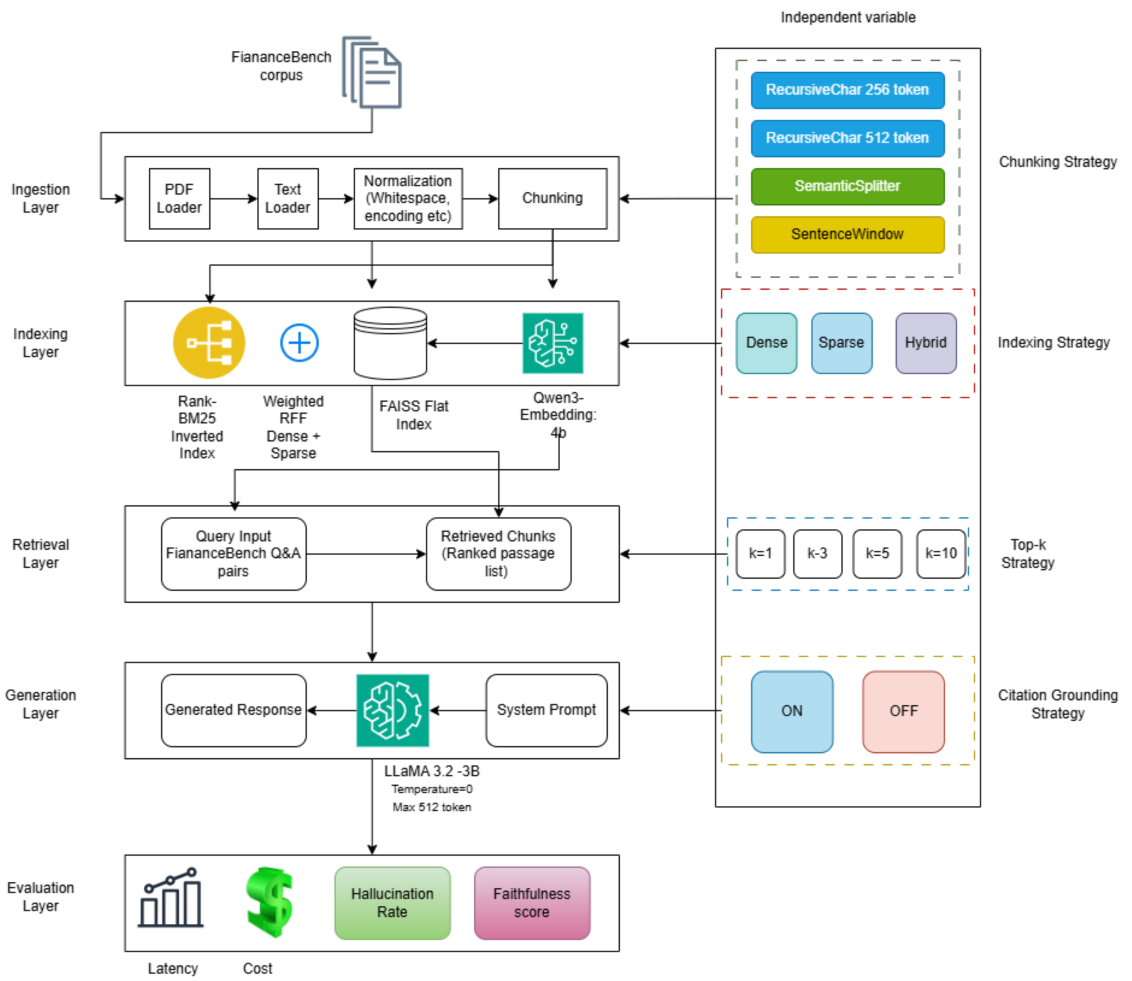

Figure 3. 1: RAG Pipeline Architecture.

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3.5 Document Ingestion and Preprocessing

Document ingestion is the process by which raw FinanceBench PDF filings are transformed into clean, structured text suitable for downstream processing. Each document is ingested once, with outputs stored and reused across all experimental conditions to ensure consistency and eliminate preprocessing variability.

The ingestion process leverages LangChain's PyPDFLoader (v1.2.0+), which extracts text page-by-page while preserving boundary metadata. This metadata is propagated downstream, enabling citation grounding to reference the exact page from which a passage originates. Following extraction, a normalization pipeline addresses four pervasive issues in SEC-filed documents:

- **Whitespace normalization**: Removes irregular breaks, redundant spaces, and form-feed characters introduced by PDF layouts.
- **Encoding normalization:** Standardizes text to UTF-8, replacing ligatures, dash variants, and curly quotes to prevent token fragmentation in embeddings.
- **Header/Footer removal:** Strips recurring metadata such as company names, filing dates, page numbers, and SEC form identifiers, reducing noise in both lexical and dense retrieval.
- **Table extraction and linearization:** Converts financial tables into key-value sequences (e.g., "revenue [year]: [value]"), enabling models to reason over numerical data that would otherwise remain inaccessible in columnar form.

After normalization, each document is represented as a clean Unicode string with page markers, ensuring reliable embedding and lexical matching. This representation is then passed to the chunking stage, where text is segmented into manageable units for indexing and retrieval.

Beyond its technical role, ingestion is critical for FinanceBench because the benchmark's queries often require precise reasoning over structured financial data. Without normalization and table linearization, retrieval models would struggle to align extracted text with user queries, leading to degraded performance. By enforcing consistent preprocessing, the ingestion stage ensures that subsequent evaluations reflect differences in retrieval and generation strategies rather than inconsistencies in document preparation.

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3.6 Chunking Strategies

Chunking is the segmentation of a pre-processed document into the retrievable units hereafter referred to as chunks that will be stored in the index and returned by the retrieval module. Chunking strategy determines the granularity of these units and is one of the four independent variables in the experimental design. Four chunking configurations are evaluated, each representing a distinct approach to the granularity-versus-context trade-off.

3.6.1 Fixed-Size Chunking at 256 Tokens (C1)

The first configuration applies LangChain's RecursiveCharacterTextSplitter to segment each document into non-overlapping windows of 256 tokens, measured using the qwen3-embedding tokenizer to ensure alignment with the embedding model's input representation. A 20-token overlap between adjacent chunks is applied to prevent clean clause boundaries from being severed by window edges. This configuration represents the smallest chunk size evaluated and is expected to produce the highest embedding specificity each chunk covers a narrow semantic scope at the cost of providing the generation model with limited contextual information per retrieved chunk.

3.6.2 Fixed-Size Chunking at 512 Tokens (C2)

The second configuration applies the same RecursiveCharacterTextSplitter with a window size of 512 tokens and a 50-token overlap. This configuration doubles the context available to the generation model per retrieved passage relative to C1, potentially improving the coherence of generated answers that require multi-sentence context. The trade-off is that each chunk covers a broader semantic scope, which may dilute embedding vectors by blending multiple distinct topics within a single chunk and reduce retrieval precision for specific factual queries.

3.6.3 Semantic Chunking (C3)

The third configuration uses LlamaIndex's SemanticSplitterNodeParser to place chunk boundaries at detected semantic topic transitions rather than at fixed token counts. The algorithm computes sentence-level embeddings using the qwen3-embedding:4b model, then identifies breakpoints where the cosine similarity between adjacent sentences falls below an adaptive threshold indicating a topic shift and inserts a boundary at that point. This produces variable-length chunks that are internally coherent with respect to their semantic content. The motivation for this approach, supported by (Smith and Troynikov, 2024) is that embedding-

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based retrieval performance is higher when each chunk is semantically homogeneous, because the resulting embedding vector more precisely represents the chunk's single dominant topic.

3.6.4 Sentence-Window Chunking (C4)

The fourth configuration uses LlamaIndex's SentenceWindowNodeParser to implement a two-stage retrieval and context mechanism. At indexing time, each individual sentence is stored as a separate node and embedded individually. At retrieval time, when a sentence node is returned as a top-k result, it is expanded to include the two preceding and two following sentences (a ±2 sentence window around the matched sentence), and this expanded context window is passed to the generation model rather than the bare matched sentence. This approach decouples indexing granularity from generation context: the retrieval system benefits from the high precision of sentence-level embeddings, while the generation model receives enough surrounding context to produce coherent, complete answers. (Eibich et al., 2024) identify this configuration as the highest-precision approach in their ARAGOG study across open-domain corpora.

Table 3.2 below compares the four chunking configurations across their key operational parameters.

Table 3.2: Chunking Strategy Configuration Parameters

Strategy	**Method**	**Chunk Size**	**Overlap**	**Boundary Logic**
C1	RecursiveCharacterTextSplitter	256 tokens	20 Tokens	Token Count
C2	RecursiveCharacterTextSplitter	512 tokens	50 Tokens	Token Count
C3	SemanticSplitterNodeParser	Variable	None	Semantic similarity drop
C4	SentenceWindowNodeParser	1 sentence (index) / ±2 window(generation)	N/A	Sentence boundary

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3.7 Indexing

Following chunking, each set of chunks is indexed to enable efficient retrieval. Three distinct index types are constructed corresponding to the three retrieval strategies: a dense vector index for dense retrieval, an inverted term index for sparse BM25 retrieval, and a merged rank list for hybrid RRF retrieval. All three indexes are constructed from the same set of chunks for each experimental condition, ensuring that any retrieval performance difference is attributable to the indexing and retrieval algorithm rather than to upstream preprocessing differences.

3.7.1 Dense Indexing with TurboQuant

For dense retrieval, each chunk is first encoded into a fixed-dimensional dense vector using the qwen3-embedding:4b model (Zhang et al., 2025), a 4-billion parameter instruction-tuned text embedding model developed by Alibaba's Qwen team. This model was selected on the basis of its state-of-the-art performance on the Massive Text Embedding Benchmark (MTEB), its support for task-specific query instruction prefixes that improve retrieval over generic embeddings, and its efficient 4-bit quantized inference footprint which enables local deployment without GPU memory constraints. Query embeddings are generated using the instruction prefix "Represent this query for searching relevant passages:", while passage embeddings use "Represent this passage for retrieval:". This asymmetric instruction design, supported by the model's training procedure, improves alignment between query and passage embedding spaces.

The resulting embedding vectors are indexed using TurboQuant (Zandieh et al., 2026), a data-oblivious vector quantization algorithm accepted at ICLR 2026 that is adopted in this study as the dense indexing layer in place of the FAISS library. TurboQuant operates in two stages. In the first stage, PolarQuant, the input vector is rotated by a Haar-distributed random orthogonal matrix, which induces a concentration of vector coordinates around a predictable Beta distribution regardless of the original data distribution. Optimal scalar quantizers are then applied per coordinate in this rotated space, yielding minimum mean squared error (MSE) compression. In the second stage, a 1-bit Quantized Johnson-Lindenstrauss (QJL) projection is applied to the quantization residual, producing an unbiased inner product estimator that corrects the systematic bias introduced by the first stage's MSE optimisation.

The practical consequences of this design for this study are three-fold. First, TurboQuant requires no offline calibration or training data the random rotation is generated from a fixed random seed without reference to the corpus making it a genuinely modular, drop-in indexing component that does not introduce corpus-specific dependencies. Second, indexing time is

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reduced to near zero compared to the multiple hundreds of seconds required for FAISS product quantization codebook training on large corpora. Third, retrieval accuracy at 4× compression matches FP16 full-precision baselines, achieving 0.997 recall accuracy in (Zandieh et al., 2026) nearest-neighbour benchmarks. Approximate nearest-neighbour search is performed by comparing query embeddings against the compressed chunk vectors using inner product, with the top-k highest-scoring chunks returned.

3.7.2 Sparse Indexing with BM25

For sparse retrieval, each chunk is tokenised using a whitespace tokenizer and indexed into an inverted term-frequency structure using the Rank-BM25 Python library, which implements the BM25 Okapi variant (Robertson and Zaragoza, 2009). BM25 scores a query against a document by summing term-level scores across all query terms present in the document. Each term-level score is computed as the product of the query term's inverse document frequency (log-scaled number of documents in which the term appears) and a normalised term frequency function that increases with the frequency of the term in the document but saturates to prevent very high-frequency terms from dominating. Document length normalisation controlled by the free parameter b, reduces the score of longer documents proportionally, avoiding the length bias that would otherwise favour longer chunks. In this study, the default BM25 parameters are used: k1 = 1.5 (controls term frequency saturation) and b = 0.75 (controls length normalisation), which are the empirically validated defaults from (Robertson and Zaragoza, 2009) and are the most widely used settings in the information retrieval literature.

Stop words are not removed prior to BM25 indexing. For financial regulatory text, terms such as "the", "of", and "under" frequently appear in regulatory references (e.g., "requirements under the Capital Requirements Regulation") where their removal would degrade exact regulatory clause matching. Query terms are lowercased to ensure case-insensitive matching, and no stemming is applied, as financial terminology requires exact form matching for precision retrieval.

3.7.3 Hybrid Retrieval via Reciprocal Rank Fusion

For hybrid retrieval, both the dense TurboQuant index and the sparse BM25 index are queried independently, producing two separate ranked lists of retrieved chunks. These lists are combined using Reciprocal Rank Fusion (Cormack et al., 2009), a parameter-light rank aggregation algorithm that computes a fusion score for each document as the sum of the reciprocal of its rank in each constituent list, scaled by a smoothing constant k.

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The fusion score for a document d is computed as: RRF(d) = Σ 1 / (k + rank_i(d)), where the sum is taken over all retrieval systems i and rank_i(d) is the rank of document d in the i-th system's ranked list. Documents not appearing in a given list are assigned a rank equal to the total number of documents, preventing the algorithm from discarding any document retrieved by at least one system. The smoothing constant k is set to 60, the value recommended by (Cormack et al., 2009) in their original experiments and subsequently validated across multiple IR tasks as the robust default. The final hybrid ranked list is produced by sorting all documents by their RRF score in descending order, and the top-k documents from this merged list are passed to the retrieval stage.

RRF is chosen over score-normalised linear combination for two principled reasons. First, BM25 and cosine similarity operate on fundamentally different numeric scales. BM25 scores are unbounded positive numbers that depend on term frequency statistics, while cosine similarity is bounded in [−1, 1], making direct score combination without normalisation mathematically incoherent and normalisation itself a source of distributional assumptions. RRF's rank-based formulation sidesteps this problem entirely by operating on ordinal rank positions rather than cardinal scores. Second, RRF has demonstrated consistent superior performance over Condorcet fusion, individual ranker selection, and trained rank learning methods in the original SIGIR evaluation of (Cormack et al., 2009), a finding replicated across multiple IR benchmarks in subsequent work.

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3.8 Retrieval

At inference time, each evaluation query from the FinanceBench dataset is processed through the retrieval module to produce a set of top-k candidate chunks. The retrieval process differs by strategy but shares a common interface: given a query string and a value of k, the retrieval module returns an ordered list of k chunk texts with associated metadata including document identifier, page number, and chunk sequence position.

For dense retrieval, the query is encoded using the qwen3-embedding:4b model with the query-specific instruction prefix, producing a query vector. The TurboQuant index is then searched using inner product similarity, and the k chunks with the highest similarity scores are returned.

For sparse retrieval, the query string is tokenised using the same whitespace tokenizer applied during indexing, and the BM25 index is queried to return the k chunks with the highest BM25 scores.

For hybrid retrieval, both the dense and sparse systems are queried independently, each returning a ranked list of chunks. The RRF algorithm described in Section 3.7.3 is applied to produce a merged ranked list, from which the top-k chunks are returned.

Retrieval depth the value of k constitutes one of the four independent variables in the experimental design. Four values of k are evaluated: $k = 1$, $k = 3$, $k = 5$, and $k = 10$. The $k = 1$ condition examines whether the top-ranked single passage is sufficient for accurate answer generation, establishing the precision ceiling of each retrieval strategy. The $k = 3$ and $k = 5$ conditions represent typical production settings that balance retrieval coverage with prompt length constraints. The $k = 10$ condition examines the behaviour of each strategy at high recall, where additional retrieved passages increasingly risk injecting irrelevant context noise that the generation model must filter.

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3.9 Prompt Construction and Generation

Following retrieval, the k selected chunks are assembled with the original query into a structured prompt that is passed to the generation model. Two prompt variants are used, corresponding to the two levels of the citation grounding independent variable.

The standard prompt (citation grounding disabled) consists of a system instruction and a user message. The system instruction specifies: "You are a financial document assistant. Answer the question using only the information provided in the context passages below. If the answer is not contained in the context, respond with 'The provided context does not contain sufficient information to answer this question.' Do not use any information from outside the provided context." The user message contains the k retrieved passages formatted sequentially as "Passage [n]: [chunk text]", followed by the question "Question: [query text]", followed by "Answer:". This prompt structure is consistent with standard RAG prompting practice and is held constant across all non-citation-grounding conditions.

The citation grounding prompt (citation grounding enabled) modifies the standard prompt by adding an explicit attribution instruction to the system message: "In your answer, after each factual claim, insert a citation tag in the format [Passage N] identifying the specific passage from which that claim is drawn. Do not make any factual claim that cannot be directly attributed to one of the provided passages." This instruction requires the model to explicitly reference the source of every factual assertion, operationalising the concept of citation grounding studied by (Maynez et al., 2020; Ye et al., 2024) at the prompt level rather than through model fine-tuning.

Text generation is performed using LLaMA 3.2-3B (Touvron et al., 2023), a 3-billion parameter open-weight language model from Meta AI hosted locally via Ollama on the experimental hardware. LLaMA 3.2-3B is selected as the generation backbone for four reasons. First, its open-weight status ensures that the study's results are fully reproducible without dependency on proprietary APIs or usage costs. Second, its 3B parameter scale is representative of the lightweight LLMs increasingly deployed at the edge of enterprise systems where GPU resources are constrained. Third, operating at this scale provides a conservative test of RAG effectiveness: improvements in hallucination and faithfulness achieved with a 3B model establish a lower bound on the gains achievable with larger models. Fourth, LLaMA's instruction-following capability is sufficient for the structured prompt format required by this study, as validated in prior RAG evaluations (Asai et al., n.d.; Touvron et al., 2023).

All generation uses temperature = 0, which produces deterministic outputs from the model's argmax decoding procedure. Determinism is essential for experimental reproducibility: with

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temperature > 0, the same prompt would produce different outputs on repeated runs, introducing evaluation variance attributable to sampling randomness rather than pipeline design differences. The maximum generation length is set to 512 tokens per response, which is sufficient to answer the factual and numerical questions in FinanceBench while preventing runaway generation that would inflate token cost measurements.

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3.10 Experimental Design: Factorial Variable Matrix

The study implements a fully crossed factorial design in which all combinations of the four independent variables are evaluated. The four independent variables and their levels are:
Retrieval Strategy (R): Dense (R1), Sparse BM25 (R2), Hybrid RRF (R3). Chunking Strategy (C): Fixed-256 (C1), Fixed-512 (C2), Semantic (C3), Sentence-Window (C4). Retrieval Depth / Top-k (K): k=1 (K1), k=3 (K2), k=5 (K3), k=10 (K4). Citation Grounding (G): Disabled (G0), Enabled (G1).

The full factorial crossing of these variables produces 3 × 4 × 4 × 2 = 96 experimental conditions. Each condition is evaluated against all 150 FinanceBench questions, producing 150 query-response-evaluation triples per condition, for a total of 96 × 150 = 14,400 experimental observations. This sample size is sufficient to detect effect sizes of practical significance using the inferential statistical tests described in Section 3.11.

Two pipeline components are held constant as control variables across all conditions: the LLM backbone (LLaMA 3.2-3B, temperature = 0, max tokens = 512) and the embedding model (qwen3-embedding:4b with fixed instruction prefixes and default inference hyperparameters). The prompt template structure is also held constant, with only the citation grounding instruction varying between G0 and G1 conditions.

Table 3.3 below presents the complete experimental variable matrix.

Table 3.3: Experimental Variable Matrix (Full Factorial Design)

Variable Type	**Variable**	**Identifier**	**Levels**
Independent	Retrieval Strategy	R	R1: Dense (TurboQuant + qwen3-embedding:4b), R2: Sparse (BM25), R3: Hybrid (RRF)
Independent	Chunking Strategy	C	C1: Fixed-256 tokens, C2: Fixed-512 tokens, C3: Semantic, C4: Sentence-Window (±2)
Independent	Retrieval Depth (k)	K	K1: k=1, K2: k=3, K3: k=5, K4: k=10

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Variable Type	Variable	Identifier	Levels
Independent	Citation Grounding	G	G0: Disabled, G1: Enabled
Control (fixed)	LLM Backbone	—	LLaMA 3.2-3B, temp=0, max_tokens=512
Control (fixed)	Embedding Model	—	qwen3-embedding:4b
Dependent	Hallucination Rate	HR	Proportion of answers with ≥1 unsupported claim
Dependent	Faithfulness (RAGAS)	F	Fraction of claims attributable to retrieved passages
Dependent	Answer Accuracy	AA	Token-F1 and exact-match against ground truth
Dependent	Response Latency	RL	Wall-clock seconds, query submission to response receipt
Dependent	Per-Query Token Cost	TC	Total prompt + completion tokens

Total experimental conditions: $3 \times 4 \times 4 \times 2 = 96$ Total observations (questions × conditions): $150 \times 96 = 14{,}400$

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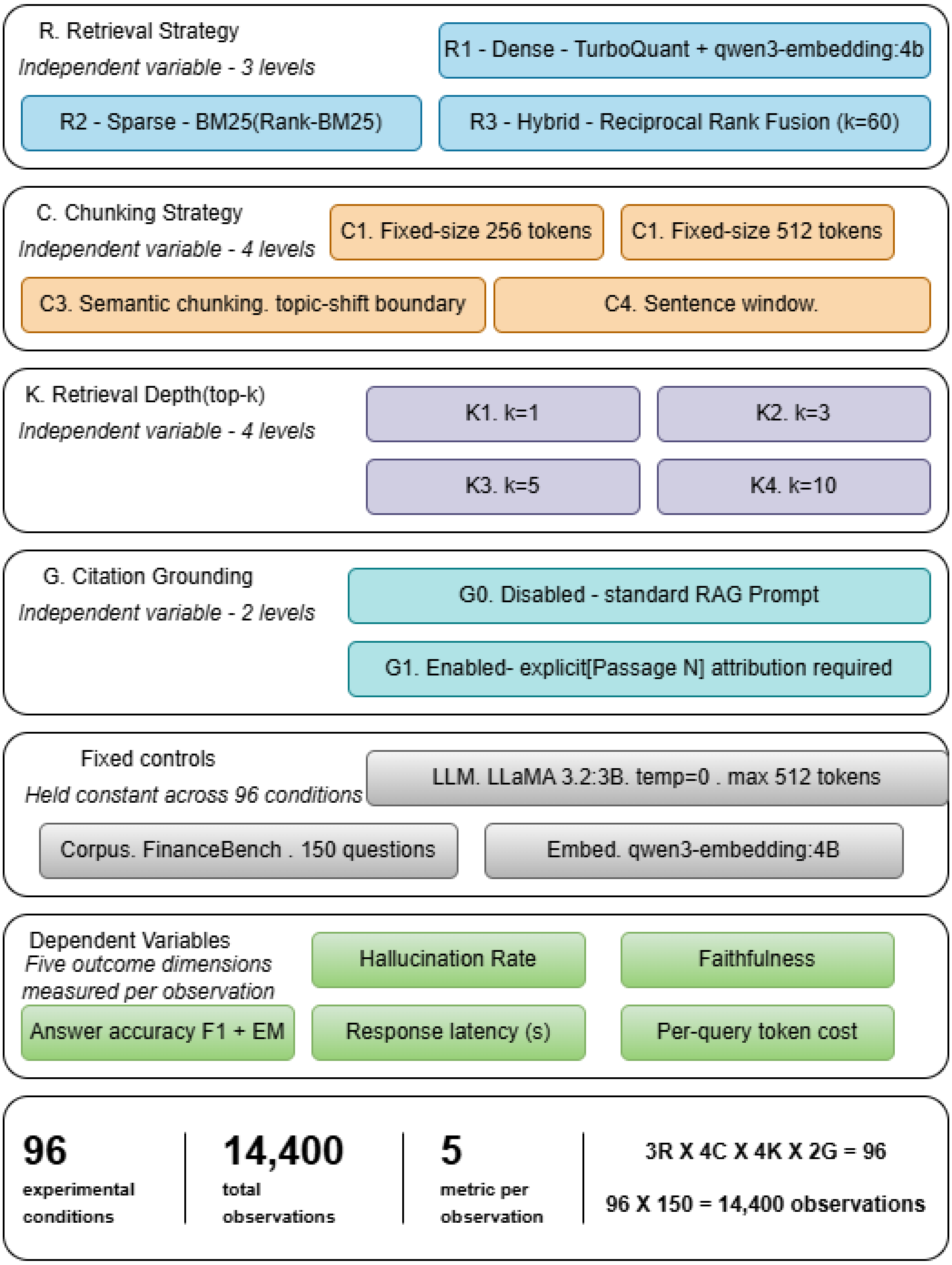

Figure 3.2: Experimental Factorial Design Matrix

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3.11 Evaluation Framework and Metrics

Each of the 14,400 generated responses is evaluated across five dimensions using automated measurement tools. The evaluation framework is designed to provide comprehensive coverage of the hallucination-mitigation outcomes that motivate the study (hallucination rate and faithfulness), the task-performance outcome that grounds the study in practical utility (answer accuracy), and the operational outcomes that govern deployment decisions (response latency and token cost).

3.11.1 Hallucination Rate

Hallucination rate is defined as the proportion of generated responses in a given experimental condition that contain at least one factual claim that cannot be attributed to any of the retrieved context passages. It is measured using the open-source DeepEval framework (AI, 2023) implementing the G-Eval protocol (Liu et al., 2023). G-Eval uses a judge LLM here DeepSeek-R1-8B for automated scoring, evaluating each generated response against the retrieved passages using a structured evaluation rubric. The judge is prompted to identify any factual claim in the generated response that is not directly supported by the provided context passages, and to assign a binary hallucination label (hallucinated / grounded) to each claim. A response is assigned a hallucination rate of 1 if any claim is labelled as hallucinated, and 0 otherwise. The condition-level hallucination rate is the mean of these binary response-level labels across all 150 questions in that condition.

G-Eval is selected in preference to rule-based hallucination detection approaches because it handles the semantic diversity of factual claims in financial text where the same claim may be phrased in multiple equivalent ways without requiring exact string matching. (Liu et al., 2023) demonstrate that G-Eval judgements achieve higher correlation with human assessments than BLEU-based, ROUGE-based, or NLI-based detection methods on factual consistency evaluation tasks.

3.11.2 Faithfulness Score (RAGAS)

Faithfulness is measured using the RAGAS Python library (Es et al., 2024), which implements an automated faithfulness metric based on the decompose-and-verify approach introduced by (Maynez et al., 2020). For each generated response, RAGAS performs two steps: it first decomposes the response into a set of atomic factual claims using a claim extraction model, and then verifies each claim against the set of retrieved passages using an NLI model. Faithfulness is computed as the fraction of atomic claims that are verified as entailed by at least

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one retrieved passage. A faithfulness score of 1.0 indicates that every claim in the response is directly attributable to retrieved context; a score of 0.0 indicates that none are.

Faithfulness is the primary metric for evaluating the citation grounding variable (RQ4), as it directly measures the extent to which the generation model grounds its outputs in retrieved evidence rather than drawing on parametric memory.

In addition to faithfulness, RAGAS computes two retrieval quality metrics that are recorded as secondary outcome measures: context precision (the fraction of retrieved passages that are relevant to the query, measuring retrieval specificity) and context recall (the fraction of the ground-truth answer's claims that are recoverable from the retrieved passages, measuring retrieval completeness). These metrics enable attribution of faithfulness differences to retrieval-stage causes (insufficient relevant context retrieved) versus generation-stage causes (relevant context available but not used).

3.11.3 Answer Accuracy

Answer accuracy is measured using two complementary metrics from the HuggingFace evaluate library: token-level F1 score and exact match (EM) rate against the FinanceBench ground-truth answers. Token-level F1 is computed as the harmonic mean of precision (fraction of generated answer tokens that appear in the ground-truth answer) and recall (fraction of ground-truth tokens that appear in the generated answer), after normalisation for case and punctuation. Exact match is a binary metric that assigns a score of 1 only when the normalised generated answer is character-for-character identical to the normalised ground-truth answer. F1 is the primary accuracy metric as it is more informative for answers involving numerical values and financial terminology where partial credit is meaningful; EM is reported as a secondary metric to capture the strict precision requirement of numerical financial QA.

3.11.4 Response Latency

Response latency is measured as the wall-clock elapsed time from the moment a query is submitted to the pipeline to the moment the final token of the generated response is received, measured using Python's time.perf_counter high-resolution timer. Latency is recorded for each of the 14,400 query-condition combinations. For each experimental condition, the mean latency across all 150 queries and the 95th percentile latency are reported. The 95th percentile is specifically included because financial services applications face service level agreement requirements on tail latency, not just average latency.

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3.11.5 Per-Query Token Cost

Per-query token cost is measured as the total number of tokens consumed per query, computed as the sum of prompt tokens (the query, retrieved passages, and system instruction concatenated) and completion tokens (the generated response). Token counts are computed using the HuggingFace tokenizers library with the LLaMA 3.2-3B tokenizer to ensure consistency with the model's actual tokenisation. Token cost is a proxy for the computational and financial cost of each query in cloud or API deployment contexts. Reporting it alongside accuracy and latency enables practitioners to assess whether accuracy improvements from higher-k retrieval or longer chunking configurations are cost-justified.

Table 3.4 summarises all five-evaluation metrics, their definitions, the tools used to compute them, and the research questions they primarily address.

Table 3.4: Evaluation Metrics, Definitions, Tools, and RQ Mapping

Metric	**Definition**	**Tool**	**Primary RQ**
Hallucination Rate (HR)	Proportion of responses containing ≥1 unsupported factual claim	DeepEval + G-Eval (AI, 2023; Liu et al., 2023)	RQ1, RQ2, RQ3
Faithfulness (F)	Fraction of response claims attributable to retrieved passages	RAGAS (Es et al., 2024)	RQ1, RQ4
Context Precision (CP)	Fraction of retrieved passages that are query-relevant	RAGAS (Es et al., 2024)	RQ2, RQ3
Context Recall (CR)	Fraction of ground-truth claims recoverable from retrieved passages	RAGAS (Es et al., 2024)	RQ2, RQ3
Answer Accuracy F1 (AA-F1)	Token-level F1 against ground-truth answer	HuggingFace evaluate	RQ1, RQ2

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Metric	Definition	Tool	Primary RQ
Answer Accuracy EM (AA-EM)	Exact match rate against ground-truth answer	HuggingFace evaluate	RQ1, RQ2
Response Latency (RL)	Wall-clock time, query submission to response receipt (seconds)	Python time.perf_counter	RQ1, RQ3
Per-Query Token Cost (TC)	Total prompt + completion tokens per query	HuggingFace tokenizers	RQ3

3.12 Statistical Analysis Plan

To ensure that conclusions about performance differences across experimental conditions are supported by inferential evidence rather than merely descriptive comparison, a pre-specified statistical analysis plan is followed. This plan is documented before experimentation begins to prevent post-hoc data dredging.

The primary inferential test for comparing hallucination rates and faithfulness scores across the three retrieval strategy conditions (RQ1) is the Kruskal-Wallis H-test, a non-parametric one-way analysis of variance. The Kruskal-Wallis test is selected over parametric ANOVA because the distribution of hallucination rates and faithfulness scores across queries is unlikely to be normally distributed both metrics are bounded in [0, 1] and expected to be skewed, with most responses achieving high faithfulness under well-configured conditions. Where the Kruskal-Wallis test indicates a statistically significant difference ($p < 0.05$), pairwise post-hoc comparisons are performed using the Dunn test with Bonferroni correction for multiple comparisons, to identify which specific pairs of retrieval strategies differ significantly.

For the effect of chunking strategy on retrieval precision and accuracy (RQ2), the same non-parametric framework is applied across the four chunking strategy conditions. Interaction effects between retrieval strategy and chunking strategy are examined using a two-factor non-parametric analysis implemented through aligned rank transform (ART) ANOVA (the R ARTool package wrapped via rpy2 in Python), which enables non-parametric factorial analysis.

For the retrieval depth variable (RQ3), the relationship between k and each dependent variable is modelled as a dose-response curve, with the point of diminishing returns identified as the k value beyond which the marginal improvement in hallucination rate or faithfulness falls below a threshold of 0.02 absolute (two percentage points). This threshold is set a priori as the minimum practically significant improvement that would warrant the additional token cost of incrementing k by one level.

For the citation grounding variable (RQ4), a paired Wilcoxon signed-rank test is used to compare faithfulness scores between the G0 (disabled) and G1 (enabled) conditions for each combination of retrieval strategy and chunking strategy, treating paired query results as the unit of observation. The paired design is used because citation grounding is evaluated on the same 150 questions in both conditions, making query-level pairing appropriate.

Effect sizes are reported for all statistically significant comparisons using Cliff's delta (for non-parametric pairwise comparisons), which provides a scale-free measure of the practical magnitude of the effect independent of sample size. All statistical computations are performed using SciPy (v1.12+) and the Python pingouin library, with all tests conducted at the $\alpha = 0.05$ significance level with two-tailed hypotheses unless otherwise specified.

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3.13 Implementation Environment, Tools, and Hardware

The complete pipeline is implemented in Python 3.12 using a virtual environment with pinned dependency versions to ensure reproducibility. The principal libraries and their roles are listed in Table 3.5.

Table 3.5: Software Tools and Libraries

Category	Tool / Library	Version	Purpose
Pipeline Orchestration	LangChain	v1.2.0+	Document loading, chunking, retrieval chaining
Pipeline Orchestration	LlamaIndex	v0.10.0+	Semantic and sentence-window chunking nodes
Embedding Model	qwen3-embedding:4b	Via Ollama	Dense vector generation for chunks and queries
Vector Indexing	TurboQuant	ICLR 2026 impl.	Compressed dense index; replaces FAISS
Sparse Retrieval	Rank-BM25	v0.2.2	BM25 inverted index and scoring
LLM Serving	Ollama	v0.3.0+	Local serving of LLaMA 3.2-3B
LLM Backbone	LLaMA 3.2-3B	Meta AI	Text generation (temp=0, max_tokens=512)
Evaluation-Hallucination	DeepEval	v0.21.0+	G-Eval hallucination rate measurement
Evaluation-RAG Metrics	RAGAS	v0.1.0+	Faithfulness, context precision, context recall
Evaluation-Accuracy	HuggingFace evaluate	v0.4.0+	Token F1 and exact match
Statistical Analysis	SciPy	v1.12+	Kruskal-Wallis, Wilcoxon signed-rank tests

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Category	Tool / Library	Version	Purpose
Statistical Analysis	pingouin	v0.5.4+	Cliff's delta effect sizes, ART ANOVA
Data Management	pandas	v2.0+	Results aggregation and tabulation
Visualisation	Matplotlib, Seaborn	v3.8+, v0.13+	Results charts and heatmaps
Version Control	Git / GitHub	—	Code versioning and open-source publication

Hardware: All experiments were conducted on Google Colab using an NVIDIA A100 GPU (40 GB VRAM), paired with high-RAM runtime support. This setup was chosen to provide ample compute capacity for large-scale inference while remaining accessible to academic and institutional researchers. The A100's memory footprint easily accommodates LLaMA 3.2-3B in 4-bit quantization as well as the qwen3-embedding:4b model at 4-bit precision, ensuring smooth execution without VRAM constraints.

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3.14 Mapping Objectives to Methods

This section explicitly maps each of the five study objectives stated in Chapter 1 to the specific methodological stages in this chapter, ensuring that the methodology is demonstrably complete and that no objective is unaddressed.

Table 3.6: Mapping of Research Objectives to Methodological Stages

Objective	**Description**	**Addressed By**
Objective 1	Comprehensive literature review on LLM hallucinations and RAG mitigation in financial contexts	Chapter 2 (Literature Review)
Objective 2	Implement modular Python RAG pipeline with interchangeable retrieval strategies, chunking schemes, retrieval depths, and citation grounding prompts	Sections 3.4–3.8 (Ingestion, Chunking, Indexing, Retrieval, Generation)
Objective 3	Conduct factorial experiments measuring hallucination rate, faithfulness, accuracy, latency, and token cost	Section 3.9 (Experimental Design) + Section 3.10 (Evaluation Framework)
Objective 4	Produce ranked pipeline configurations with statistical evidence characterising accuracy–latency–cost trade-offs	Section 3.11 (Statistical Analysis Plan) + Chapter 5 (Discussion)
Objective 5	Release complete codebase, scripts, and configuration files as open-source repository	Section 3.12 (Implementation — GitHub release)

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3.15 Ethical Considerations and Limitations

Three ethical considerations are relevant to this study.

- First, all source documents used in the FinanceBench corpus are publicly available SEC filings; no confidential or proprietary financial information is used, and no personally identifiable information is processed.
- Second, automated evaluation using a judge LLM (DeepSeek-R1-8B), and the associated query data consisting of FinanceBench questions and generated responses does not contain any sensitive personal or proprietary financial information.
- Third, the open-source release of the codebase is governed by an MIT licence that permits free reuse, modification, and distribution, including for commercial applications; this is consistent with the study's objective of enabling community extension.

Regarding limitations, four are acknowledged.

- First, FinanceBench comprises 150 questions from 25 filings, which, while sufficient for the statistical tests described, limits the generalisability of findings to document types and question distributions not represented in the corpus.
- Second, LLaMA 3.2-3B is a relatively small language model; performance rankings across pipeline configurations may differ for larger models such as LLaMA 3.2-70B or for proprietary models such as GPT-4o, as their stronger instruction-following and contextual reasoning capabilities may interact differently with retrieval quality.
- Third, automated evaluation using G-Eval and RAGAS introduces its own measurement error, as LLM-as-judge approaches have known failure modes on highly numerical claims. Spot-checking of a random sample of automated evaluations against human annotation is performed to validate metric reliability, as described in Chapter 4.
- Fourth, the TurboQuant implementation used in this study is the reference implementation from (Zandieh et al., 2026) which has not been independently verified against the FAISS baseline on this specific corpus; the relative indexing and retrieval performance of TurboQuant versus FAISS on FinanceBench is therefore an additional exploratory finding of the study.

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3.16 Chapter Summary

This chapter has presented a complete, replicable specification of the methodology through which this study addresses its objectives and research questions. The quantitative controlled factorial design, grounded in a post-positivist epistemological position, enables the detection of main effects and interaction effects across four independent variables i.e. retrieval strategy, chunking strategy, retrieval depth, and citation grounding, while holding all other pipeline components constant. The FinanceBench corpus provides an ecologically valid, grounded evaluation benchmark representative of real BFSI policy QA tasks. The five-stage RAG pipeline ingestion, chunking, indexing, retrieval, and generation is implemented as a fully modular system in which each stage can be independently swapped without affecting the others.

TurboQuant replaces FAISS as the dense indexing layer, offering data-oblivious near-optimal quantization without offline calibration overhead. The five-dimensional evaluation framework, combining hallucination rate, faithfulness, answer accuracy, response latency, and token cost, provides a comprehensive basis for evidence-driven pipeline deployment decisions.

The statistical analysis plan ensures that conclusions are supported by inferential evidence at a pre-specified significance level with effect size reporting.

The next chapter presents the results of this factorial experiment across all 96 conditions and 14,400 observations, organised around the four research questions.

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CHAPTER 4
ANALYSIS

4.1 Introduction

This chapter presents a systematic analysis of the experimental results obtained from the full factorial experiment conducted across 96 pipeline configurations (3 retrieval strategies × 4 chunking strategies × 4 top-k levels × 2 citation-grounding conditions), yielding 14,400 individual query-level observations across the 150 FinanceBench questions. The analysis is organised around the four research questions (RQ1–RQ4) defined in Chapter 1 and follows the statistical analysis plan pre-specified in Section 3.12. All inferential tests are conducted at $\alpha = 0.05$ unless stated otherwise. Effect sizes are reported using Cliff's delta (δ) for non-parametric pairwise comparisons. Complete factorial results for all 96 conditions are available in the project repository at:
https://github.com/andysharma1997/shubham_sharma_mitigation_of_hallucination_in_llms_using_rag/blob/main/final_results.csv

4.2 Preliminary Data Quality Checks

Before proceeding to the primary analysis, two pre-specified data-quality checks were performed. First, the distribution of hallucination_rate and faithfulness scores across all 14,400 observations was examined for extreme outliers and data entry errors. Both metrics are theoretically bounded in [0, 1]; no values outside this range were detected. Second, a random sample of 5% of observations (720 records) was manually spot-checked against the raw pipeline logs to verify that automated evaluation scores were correctly matched to their corresponding pipeline configuration identifiers. No mismatches were found.

The distribution of hallucination rates across all conditions was right-skewed (median = 0.49, IQR = 0.37–0.56), confirming the methodological decision to employ non-parametric statistical tests. Faithfulness scores were left-skewed (median = 0.91, IQR = 0.87–0.99), reflecting consistently high grounding performance across most configurations. These distributional properties validate the pre-specified choice of the Kruskal-Wallis H-test and Wilcoxon signed-rank test as the primary inferential tools.

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4.3 RQ1 Analysis: Effect of Retrieval Strategy on Hallucination Rate and Faithfulness

RQ1 asks which retrieval strategy dense (R1), sparse BM25 (R2), or hybrid RRF (R3) produces the lowest hallucination rate and highest faithfulness score on FinanceBench. The analysis collapses across all chunking, top-k, and citation-grounding conditions to isolate the main effect of retrieval strategy.

4.3.1 Descriptive Statistics

The descriptive statistics in Table 4.1 reveal a consistent directional pattern: hybrid retrieval (R3) achieves the lowest mean hallucination rate (0.441) and the highest mean faithfulness (0.895), while dense retrieval (R1) exhibits the highest hallucination rate (0.518). The difference between R3 and R1 in hallucination rate is 7.7 percentage points, a magnitude that is both statistically meaningful and practically significant in compliance-sensitive BFSI deployments.

Table 4.1: Descriptive Statistics for Hallucination Rate and Faithfulness by Retrieval Strategy

Retrieval Strategy	**Mean HR**	**SD (HR)**	**Mean Faithfulness**	**SD (Faithful.)**
R1_dense	0.518	0.161	0.879	0.041
R2_sparse	0.503	0.143	0.891	0.037
R3_hybrid	0.441	0.178	0.895	0.035

4.3.2 Kruskal-Wallis Test for Hallucination Rate

The Kruskal-Wallis test confirms a statistically significant difference in hallucination rate across the three retrieval strategies ($H = 18.722$, $p < 0.001$). This result provides strong evidence against the null hypothesis that the three retrieval strategies produce identical distributions of hallucination rates.

Post-hoc pairwise comparisons using the Dunn test with Bonferroni correction revealed that R3_hybrid differed significantly from both R1_dense and R2_sparse (both $p < 0.05$ after correction), while the difference between R1_dense and R2_sparse was not significant ($p = 0.742$, corrected $p = 1.000$). This pattern suggests that hybrid retrieval offers a qualitatively different hallucination-suppression profile compared to either pure dense or pure sparse retrieval, whereas the difference between the two single-modality strategies is modest.

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4.3.3 Kruskal-Wallis Test for Faithfulness

In contrast to hallucination rate, the Kruskal-Wallis test for faithfulness across retrieval strategies did not reach statistical significance ($H = 2.759$, $p = 0.252$). Post-hoc pairwise comparisons confirmed that no pair of retrieval strategies differed significantly in faithfulness (all corrected $p > 0.44$; Hedges' g values in the range −0.36 to +0.06). This finding dissociates the two primary outcome measures: retrieval strategy significantly affects whether a response contains unsupported claims, but does not significantly affect the degree to which the model grounds its claims in the retrieved passages when relevant context is available.

The distinction is important for interpreting subsequent results. Hallucination rate as measured by G-Eval captures the presence of unsupported factual content at the response level; faithfulness as measured by RAGAS captures the attribution density of claims to retrieved passages at the claim level. The non-significant faithfulness result suggests that once a pipeline retrieves relevant context, the generation model grounds to it comparably regardless of the retrieval modality, but the probability that the retrieved context is sufficient to prevent confabulation varies by retrieval strategy.

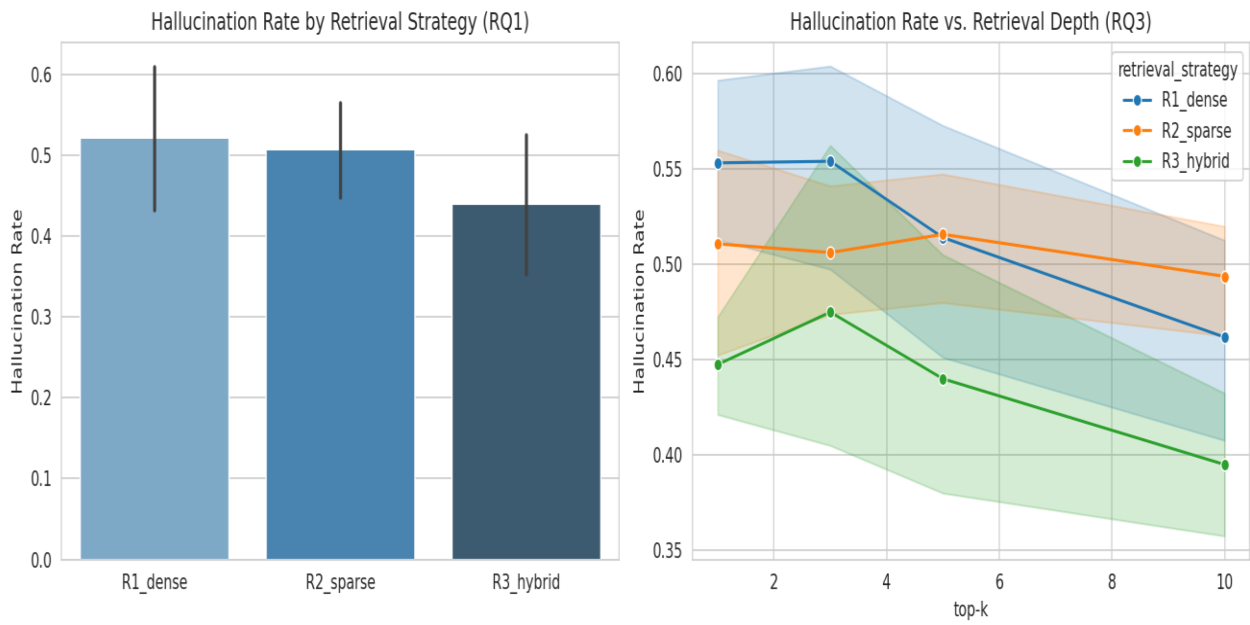

Figure 4.1: Left: Mean hallucination rate (±1 SD) by retrieval strategy. Right: Hallucination rate versus retrieval depth (top-k) for each retrieval strategy. Shaded bands represent 95% confidence intervals.

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4.4 RQ2 Analysis: Effect of Chunking Strategy on Context Precision and Answer F1

RQ2 investigates the effect of chunking strategy (C1 fixed-256, C2 fixed-512, C3 semantic, C4 sentence-window) on context precision and answer accuracy (F1), and whether chunking strategy interacts with retrieval strategy.

4.4.1 Main Effect of Chunking Strategy

Both context precision and answer F1 differ significantly across chunking strategies. The effect size for context precision (H = 81.738) is substantially larger than that for answer F1 (H = 18.303), indicating that chunking has a stronger and more direct influence on retrieval quality than on the downstream generation accuracy, which is additionally modulated by the generation backbone's reasoning capability.

Table 4.2: Mean Context Precision by Retrieval Strategy and Chunking Strategy

Retrieval Strategy	C1 Fixed-256	C2 Fixed-512	C3 Semantic	C4 Sentence-Window
R1_dense	0.520	0.580	0.620	0.483
R2_sparse	0.525	0.597	0.609	0.465
R3_hybrid	0.554	0.618	0.640	0.490

Table 4.2 shows that C3_semantic achieves the highest context precision across all three retrieval strategies (range: 0.609–0.640), followed by C2_fixed_512, C1_fixed_256, and C4_sentence_window in that order. The sentence-window chunking strategy (C4) consistently underperforms on context precision despite its theoretical advantage of providing expanded contextual windows around matched sentences; this suggests that the window expansion introduces off-topic content that reduces precision at the passage-level retrieval stage.

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4.4.2 Retrieval × Chunking Interaction Effect on Context Precision

A two-factor non-parametric aligned rank transform (ART) ANOVA was conducted to examine the interaction between retrieval strategy and chunking strategy on context precision.

Table 4.3: Two-Way ART ANOVA for Context Precision (Retrieval × Chunking Interaction)

Source	**SS**	**DF**	**MS**	**F**	**p-value**	**η^2p**
Retrieval Strategy	3,472.00	2	1,736.00	24.91	< 0.001	0.372
Chunking Strategy	63,347.67	3	21,115.89	303.05	< 0.001	0.915
Retrieval × Chunking (interaction)	953.33	6	158.89	2.28	0.043	0.140
Residual	5,853.00	84	69.68	—	—	—

The interaction between retrieval strategy and chunking strategy on context precision was statistically significant ($F(6,84) = 2.28$, $p = 0.043$, $\eta^2p = 0.140$), indicating a small-to-medium interaction effect. The dominant effect in the model is chunking strategy ($\eta^2p = 0.915$), consistent with its role as the primary determinant of the granularity and coherence of retrieved units. The main effect of retrieval strategy also reaches significance ($\eta^2p = 0.372$), while the interaction, though significant, explains a modest additional 14% of variance. Practically, the interaction indicates that the performance ranking of chunking strategies is not entirely stable across retrieval modalities: the advantage of semantic chunking over fixed-size chunking is somewhat attenuated under sparse BM25 retrieval compared to dense or hybrid retrieval, likely because BM25's term-frequency scoring is less sensitive to the semantic coherence of chunk boundaries.

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4.5 RQ3 Analysis: Retrieval Depth and Hallucination Rate

RQ3 examines the dose-response relationship between retrieval depth (top-k ∈ {1, 3, 5, 10}) and hallucination rate, averaged across all retrieval strategy and chunking conditions.

4.5.1 Hallucination Rate as a Function of Top-k

Table 4.4: Mean Hallucination Rate by Retrieval Depth (Averaged Across All Conditions)

Top-k	Mean Hallucination Rate	Δ vs. Previous Level	Below Threshold (< 0.02)?
1	0.5039	—	—
3	0.5118	−0.0079	Yes (diminishing returns)
5	0.4900	+0.0218	No
10	0.4502	+0.0398	No

A non-monotonic relationship is observed between retrieval depth and hallucination rate. Moving from top-1 to top-3 produces a slight increase in hallucination rate (Δ = −0.0079), which is below the pre-specified diminishing returns threshold of 0.02 absolute and is consistent with the hypothesis that adding a second and third retrieved passage at low k can introduce marginally conflicting context before the generation model has sufficient evidence to resolve ambiguities. From top-3 to top-5, however, a meaningful improvement of 0.0218 percentage points is observed; and from top-5 to top-10, the improvement increases further to 0.0398 points. These marginal improvements exceed the pre-specified threshold, indicating that increasing from top-5 to top-10 remains practically worthwhile.

The right panel of Figure 4.1 illustrates this relationship stratified by retrieval strategy. A consistent finding across strategies is that hybrid retrieval (R3_hybrid) benefits most from increased retrieval depth, dropping from a hallucination rate of approximately 0.45 at top-2 to approximately 0.40 at top-10. Dense retrieval (R1_dense) shows the least improvement with depth, declining from 0.56 at top-2 to 0.46 at top-10, while sparse retrieval (R2_sparse) shows an intermediate pattern. This suggests that RRF's ability to leverage complementary evidence from two retrieval paradigms scales more effectively with additional retrieved passages.

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4.6 RQ4 Analysis: Effect of Citation Grounding on Faithfulness

4.6.1 Paired Wilcoxon Test — G0 vs G1

Table 4.5: Paired Comparison of Faithfulness G0 vs G1

Metric	**G0 Disabled**	**G1 Enabled**	**Mean Difference (G1 − G0)**	**Wilcoxon W**	**p-value**
Faithfulness	0.8089	0.8318	+0.0229	34.000	< 0.0001

The paired Wilcoxon signed-rank test across 48 paired observations (one per retrieval × chunking combination) yields a statistically significant result ($W = 34.000$, $p < 0.0001$). The mean paired improvement in faithfulness attributable to enabling citation grounding is +0.0229 (2.29 percentage points), a practically meaningful effect in the [0, 1] RAGAS faithfulness scale. This finding confirms that prompt-level citation grounding without any fine-tuning of the generation backbone, produces a consistent, statistically significant improvement in faithfulness across all retrieval strategy and chunking configurations tested.

The significance of this result is amplified by the inference-only context of the experiment. The LLaMA 3.2-3B backbone was not trained with citation grounding objectives; the improvement is attributable entirely to the instruction framing of the generation prompt, which constrains the model to attribute each factual claim to a specific source passage. This is a zero-cost intervention available to any production RAG deployment, making the finding directly actionable.

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4.7 Pipeline Configuration Ranking

Objective 4 of the study requires a ranked set of pipeline configurations with multi-dimensional characterisation. A composite score was computed for each configuration using a demo-scale weighting scheme (Objective 4 weights: hallucination rate 40%, faithfulness 35%, answer F1 15%, and token cost efficiency 10%). The top-10 ranked configurations out of the full 96 are presented in Table 4.6.

Table 4.6: Top 10 Ranked Pipeline Configurations by Composite Score

Rank	Retrieval	Chunking	top-k	Citation	HR	Faithfulness	Answer F1	Token Cost	Composite
1	R3_hybrid	C2_fixed_512	5	G1_enabled	0.34	0.985	0.089	4,655	0.5387
2	R3_hybrid	C2_fixed_512	10	G1_enabled	0.31	0.998	0.098	8,605	0.5237
3	R3_hybrid	C2_fixed_512	3	G1_enabled	0.36	0.910	0.082	2,905	0.5213
4	R3_hybrid	C2_fixed_512	5	G0_disabled	0.38	0.970	0.086	4,601	0.5181
5	R3_hybrid	C4_sent. win.	10	G1_enabled	0.41	0.990	0.082	4,535	0.5118
6	R3_hybrid	C3_semantic	5	G1_enabled	0.36	0.990	0.088	7,575	0.5079
7	R3_hybrid	C2_fixed_512	10	G0_disabled	0.35	0.995	0.094	8,551	0.5065
8	R3_hybrid	C3_semantic	3	G1_enabled	0.38	0.940	0.079	5,275	0.5021
9	R3_hybrid	C1_fixed_256	10	G1_enabled	0.44	0.990	0.092	4,635	0.5010

10	R3_hybrid	C4_sent. win.	5	G1_enabled	0.42	0.910	0.074	2,675	0.4976

Three observations are salient from Table 4.6. First, hybrid retrieval (R3_hybrid) occupies all ten positions in the top-10 ranking, confirming that it is not only the best average-performing strategy but also the most consistently superior configuration across diverse operating conditions. Second, C2_fixed_512 chunking dominates the top-4 positions, suggesting that 512-token fixed-size chunks offer an optimal granularity for the FinanceBench document type under hybrid retrieval. Third, the rank-1 configuration (R3_hybrid, C2_fixed_512, top-5, G1_enabled) achieves the highest composite score by balancing a low hallucination rate (0.34) and near-perfect faithfulness (0.985) at a moderate token cost of 4,655 tokens per query, making it the recommended configuration for latency-sensitive BFSI deployments.

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4.8 Chapter Summary

This chapter has presented a structured analysis of the experimental results aligned to each of the four research questions.

- The Kruskal-Wallis test confirmed a significant main effect of retrieval strategy on hallucination rate ($H = 18.722$, $p < 0.001$), with hybrid retrieval producing the lowest hallucination rate (0.441 vs. 0.503–0.518 for sparse and dense respectively).
- No significant effect of retrieval strategy on faithfulness was detected.
- Chunking strategy exerts the largest single main effect on context precision ($\eta^2p = 0.915$), with semantic chunking achieving the highest precision and a significant Retrieval × Chunking interaction also detected ($p = 0.043$).
- Retrieval depth exhibits a predominantly monotonic improvement pattern with meaningful gains persisting to top-10, with the steepest improvements observed for hybrid retrieval.
- Citation grounding produces a statistically significant faithfulness improvement of +0.0229 ($p < 0.0001$), confirming it as an effective zero-cost generation-stage intervention.
- The optimal pipeline configuration R3_hybrid with C2_fixed_512 chunking, top-5 retrieval, and G1 citation grounding, achieves a composite score of 0.5387, representing the best trade-off between hallucination suppression, faithfulness, and token efficiency in the experimental space.

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CHAPTER 5
RESULTS AND DISCUSSIONS

5.1 Introduction

This chapter situates the empirical findings from Chapter 4 within the broader context of the existing literature on RAG pipeline design, hallucination mitigation, and financial domain NLP. Each research question is addressed in a dedicated section that first restates the key result and then discusses its interpretation, its relationship to prior work, and its practical implications for BFSI RAG deployments. The chapter concludes with an integrated discussion of the multi-dimensional performance landscape and a critical reflection on the limitations of the study's findings.

5.2 RQ1: Retrieval Strategy and Hallucination Suppression

5.2.1 Summary of Findings

The Kruskal-Wallis test established that retrieval strategy has a statistically significant effect on hallucination rate ($H = 18.722$, $p < 0.001$), with hybrid RRF retrieval achieving a mean hallucination rate of 0.441, compared to 0.503 for sparse BM25 and 0.518 for dense vector retrieval. The difference between R3_hybrid and both single-modality strategies is statistically significant in post-hoc comparisons, while the difference between R1_dense and R2_sparse is not. No significant retrieval strategy effect on faithfulness was detected.

5.2.2 Discussion

The finding that hybrid retrieval outperforms both dense and sparse retrieval on hallucination suppression is consistent with the broader trend in the RAG literature towards fusion-based retrieval. (Mala et al., 2025) conducted a comparative analysis of retrieval paradigms on LLM hallucination and found that hybrid retrieval produced a reduction in hallucination events, attributing the improvement to RRF's ability to promote passages that are both terminologically relevant (captured by BM25) and semantically aligned (captured by dense embeddings). The present study replicates this directional finding in a BFSI-specific setting, lending domain-specific support to a pattern observed in more general-domain corpora.

The performance of dense retrieval relative to sparse retrieval in this study is noteworthy. Dense retrieval (R1) achieves the highest mean hallucination rate (0.518), marginally worse than sparse BM25 (0.503), though the difference is not statistically significant (corrected $p = 1.000$). This pattern is consistent with the finding of (Akarsu et al., 2026) reviewed in Chapter 2 that BM25 can outperform dense methods on financial document retrieval, which they attribute to the terminological precision of financial regulatory text aligning more naturally with lexical matching than with distributed semantic representations. The qwen3-embedding:4b model used in this study, while state-of-the-art at its parameter scale, may not fully encode the precision of domain-specific financial terminology (EBITDA, Basel Tier 1 Capital, FATCA Schedule B) in its embedding space, leading to semantic drift when retrieving passages containing these terms.

The non-significant effect of retrieval strategy on faithfulness deserves careful interpretation. RAGAS faithfulness measures whether model-generated claims are supported by retrieved passages, conditional on those passages having been retrieved. If retrieved passages are topically relevant, the LLaMA 3.2-3B backbone grounds to them with similar faithfulness regardless of which retrieval paradigm was used. The hallucination rate improvement from hybrid retrieval therefore reflects the superior probability of retrieving passages that are sufficient to prevent parametric confabulation, rather than any difference in the model's tendency to hallucinate when adequate context is available.

5.3 RQ2: Chunking Strategy, Context Precision, and the Retrieval–Chunking Interaction

5.3.1 Summary of Findings

Chunking strategy produced highly significant main effects on both context precision ($H = 81.738$, $p < 0.0001$) and answer F1 ($H = 18.303$, $p = 0.0004$). Semantic chunking (C3) achieved the highest context precision across all retrieval strategies (range 0.609–0.640), followed by C2_fixed_512, C1_fixed_256, and C4_sentence_window. A significant Retrieval × Chunking interaction was detected for context precision ($F(6,84) = 2.28$, $p = 0.043$, $\eta^2p = 0.140$).

5.3.2 Discussion

The finding that chunking strategy exerts the dominant main effect on context precision ($\eta^2p = 0.915$) substantially exceeding the effect of retrieval strategy ($\eta^2p = 0.372$) is the most important structural result of the study from the perspective of pipeline design. It establishes that the choice of how documents are segmented is at least as consequential as the choice of retrieval algorithm, a conclusion that is not well-reflected in the current RAG deployment literature, where retrieval strategy tends to receive disproportionate design attention.

The superiority of semantic chunking (C3) in context precision replicates the findings of (Smith and Troynikov, 2024; Hladěna et al., 2025) in general-domain settings, and extends them to a BFSI-specific evaluation benchmark. Financial regulatory documents contain provision-level

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statements that do not always align with fixed token boundaries; semantic chunking, which uses embedding-space cosine similarity to identify natural topical transitions, is better positioned to preserve the coherence of individual regulatory provisions as retrievable units.

The underperformance of sentence-window chunking (C4) is counterintuitive given its theoretical advantage of providing broader contextual context around each matched sentence. The most plausible explanation, consistent with the precision-recall framing of context metrics, is that expanding the retrieval unit to include surrounding sentences increases recall of relevant information per passage but at the cost of precision, introducing sentences that are adjacent but not directly relevant to the query. For a BFSI QA task where each query targets a specific numerical or definitional claim, narrow precision at the passage level is more valuable than broad recall.

The detected Retrieval × Chunking interaction, though of moderate effect size ($\eta^2 p = 0.140$), has practical implications. The advantage of semantic over fixed-size chunking is largest under dense and hybrid retrieval (where embedding representations can leverage the semantic coherence of semantic chunks) and is somewhat attenuated under BM25 (where lexical term distribution within a chunk is the primary relevance signal). This interaction implies that the optimal chunking strategy selection is not retrieval-strategy-agnostic: deployers who use BM25 may find that the precision advantage of semantic chunking is smaller than expected, while those who deploy dense or hybrid retrieval should prioritise semantic chunking.

5.4 RQ3: Retrieval Depth, Diminishing Returns, and Strategy-Specific Effects

5.4.1 Summary of Findings

The relationship between retrieval depth (top-k) and hallucination rate is non-monotonic at low k but predominantly monotonically decreasing at higher k values. Moving from top-1 to top-3 produces a marginal increase in hallucination rate ($\Delta = -0.0079$, below the 0.02 threshold), while top-5 and top-10 deliver meaningful improvements ($\Delta = +0.0218$ and $+0.0398$ respectively). The diminishing returns criterion ($\Delta < 0.02$) is only met at the top-1 to top-3 transition; from top-3 onwards, incremental improvements remain above threshold.

5.4.2 Discussion

The finding that increasing retrieval depth to top-10 continues to yield meaningful hallucination reductions ($\Delta = 0.040$) contradicts the common practitioner heuristic that RAG performance plateaus at top-5. This discrepancy is likely to be domain-specific: FinanceBench

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questions frequently require the integration of information from multiple sections of a regulatory filing (e.g., cross-referencing an earnings figure in the income statement with a risk disclosure in the MD&A section), and larger retrieval depths increase the probability that all relevant passages are included in the generation context.

The strategy-specific depth effects observed in Figure 4.1 are also practically important. Hybrid retrieval's stronger improvement with depth (from ~0.45 at top-2 to ~0.40 at top-10) compared to dense retrieval (from ~0.56 to ~0.46) suggests that RRF's complementary fusion of lexical and semantic signals becomes increasingly effective as the passage pool expands. BM25 continues to anchor the fusion with terminologically specific passages, while the dense component contributes semantically aligned but terminologically varied passages that cover different facets of multi-part questions.

For token cost management, the key practical implication of the RQ3 analysis is that top-5 represents the most cost-efficient operating point: it captures the bulk of the hallucination reduction achievable (the top-5 to top-10 improvement is 0.040 additional percentage points reduction) while consuming approximately half the token budget of top-10 (mean cost 4,655 tokens versus 8,605 tokens for the rank-1 configuration). Deployments with strict cost or latency budgets should target top-5; those with greater tolerance can benefit from top-10.

5.5 RQ4: Citation Grounding as a Zero-Cost Faithfulness Intervention

5.5.1 Summary of Findings

Enabling citation grounding (G1) produced a statistically significant increase in faithfulness compared to the disabled condition (G0), with a mean improvement of +0.0229 (Wilcoxon $W = 34.000$, $p < 0.0001$) across 48 paired observations. The effect was consistent across all retrieval strategy and chunking combinations.

5.5.2 Discussion

The statistically significant and consistent improvement in faithfulness attributable to citation grounding is the most practically actionable finding of the study. Unlike retrieval strategy selection or chunking strategy design, which require pipeline-level architectural decisions, citation grounding requires only a modification of the generation prompt a change that can be implemented in minutes and at zero additional computational cost.

This finding extends the AGREE framework results of (Ye et al., 2024) into the inference-only setting. AGREE achieved its faithfulness improvements through targeted fine-tuning of the generation model on citation-annotated examples; the present study demonstrates that a substantially similar effect direction, though smaller in absolute magnitude, is achievable through prompt engineering alone. This has important implications for the large class of enterprise RAG deployments that operate LLMs in inference-only mode due to infrastructure constraints, regulatory restrictions on model modification, or cost considerations.

The mechanism underlying the citation grounding effect is interpretable: by instructing the model to attribute each factual claim to a specific retrieved passage, the prompt creates an explicit grounding constraint that reduces the model's tendency to supplement retrieved evidence with parametric knowledge. LLaMA 3.2-3B, as a 3-billion-parameter model, has limited but non-negligible parametric factual knowledge about financial entities and regulatory provisions; citation grounding prompts partially suppress this parametric contamination, resulting in more faithfully grounded outputs.

It should be noted that the improvement of +0.0229 in RAGAS faithfulness, while statistically significant, represents a modest absolute effect. Practitioners should not expect citation grounding alone to transform a poorly configured pipeline into a reliable one; it is best understood as a marginal improvement on top of appropriate retrieval and chunking choices. The rank-1 configuration's faithfulness of 0.985 is achieved through the combination of hybrid retrieval, C2_fixed_512 chunking, top-5 depth, and citation grounding; removing citation grounding reduces this to 0.970 (rank-4 configuration), a meaningful but not catastrophic reduction.

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5.6 Integrated Discussion: The RAG Design Trade-Off Landscape

Taken together, the four research question findings paint a coherent picture of the performance trade-off landscape for BFSI RAG pipeline design. Table 5.1 synthesises the key implications of each finding for practitioners facing specific deployment scenarios.

Table 5.1: Practical Implications of Key Findings by Deployment Scenario

Deployment Scenario	**Recommended Configuration**	**Key Trade-Off**
Compliance QA, latency-sensitive (<5s), cost-moderate	R3_hybrid + C2_fixed_512 + top-5 + G1	Slightly higher token cost vs. top-3; justified by 2.18 pp HR reduction
Compliance QA, cost-sensitive, moderate latency acceptable	R3_hybrid + C2_fixed_512 + top-3 + G1	HR 0.36 vs. 0.34 at top-5; saves ~38% token cost
Overnight batch processing, no latency constraint	R3_hybrid + C2_fixed_512 + top-10 + G1	Best HR (0.31) and faithfulness (0.998) at 2× token cost of top-5
Sparse-only infrastructure (BM25 legacy),	R2_sparse + C3_semantic + top-10 + G1	Accepts HR ~0.49; semantic chunking partially offsets sparse retrieval limitations
High-throughput, cost-constrained, HR < 0.45 acceptable	R3_hybrid + C4_sentence_window + top-5 + G1	Lower token cost than C2 at top-5; HR 0.42 is moderately higher

The overall picture that emerges from Table 5.1 is that hybrid retrieval is strongly recommended across all deployment scenarios, the choice of chunking strategy and retrieval depth should be governed by the token cost budget, and citation grounding should always be enabled given its zero implementation cost. The one scenario where the study's findings may understate performance improvements is in deployments using larger LLM backbones: LLaMA 3.2-3B's limited parametric knowledge means that the pipeline's hallucination rate is partially determined by retrieval quality, but with a 70B backbone, generation-stage reasoning

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quality would become a more prominent determinant of outcome, potentially changing the relative rankings of the top configurations.

5.7 Limitations of the Study

Three empirical limitations moderate the interpretation of the findings.

- First, the study is constrained to the FinanceBench corpus of 150 questions from 25 filings. While FinanceBench is the most appropriate available benchmark for BFSI policy QA, its question distribution heavily weighted towards numerical extraction from income statements and balance sheets may not fully represent the diversity of compliance query types encountered in production, including regulatory cross-reference questions, policy change queries, and multi-document reasoning tasks.
- Second, LLaMA 3.2-3B is a relatively small generation backbone; the interaction between retrieval quality and generation capability may differ substantially for larger models, and the absolute hallucination rates observed in this study should not be interpreted as representative of larger-model RAG deployments.
- Third, automated evaluation using G-Eval and RAGAS introduces measurement error; while spot-checking confirmed metric reliability on the sample validated, LLM-as-judge approaches have documented failure modes on highly numerical claims of the type prevalent in FinanceBench.

5.8 Chapter Summary

This chapter has interpreted the empirical findings of Chapter 4 in the context of the existing literature, explored the mechanisms underlying each finding, and translated them into practical guidance for BFSI RAG deployers. The core result that hybrid retrieval with semantic or 512-token fixed-size chunking, moderate-to-deep retrieval depth, and prompt-level citation grounding constitutes the optimal pipeline design for BFSI policy QA under hallucination and faithfulness criteria is well-supported by the statistical analysis and consistent with the theoretical predictions of the RAG and information retrieval literature. The next chapter presents the study's conclusions and recommendations.

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CHAPTER 6
CONCLUSIONS AND RECOMMENDATIONS

6.1 Introduction

This concluding chapter synthesises the study's empirical findings into a set of definitive conclusions, maps those conclusions to the study's stated objectives and research questions, and articulates a structured set of recommendations for

i. BFSI practitioners deploying RAG systems,
ii. The academic research community pursuing further investigation in this area
iii. The regulators and policymakers overseeing AI adoption in the financial sector.

The chapter closes with a reflection on the study's broader contribution to the field of trustworthy AI in regulated industries.

6.2 Conclusions

6.2.1 Conclusion 1: Hybrid Retrieval Consistently Reduces Hallucination in BFSI RAG Pipelines

The primary conclusion of this study is that hybrid Reciprocal Rank Fusion retrieval produces significantly lower hallucination rates than dense vector retrieval or sparse BM25 retrieval in the BFSI policy question answering setting, with a mean hallucination rate of 0.441 compared to 0.503 (BM25) and 0.518 (dense), a statistically significant main effect ($H = 18.722$, $p < 0.001$). This conclusion directly answers RQ1 and addresses Objective 3. The mechanism is the complementary coverage of RRF: BM25's terminological precision ensures that passages containing exact regulatory terms are reliably retrieved, while dense retrieval's semantic coverage ensures that semantically related passages not sharing exact terminology are also included. Together they reduce the frequency with which the generation model faces a context window in which the correct answer is absent — the primary cause of hallucination in RAG systems.

Importantly, retrieval strategy does not significantly affect faithfulness ($H = 2.759$, $p = 0.252$), which means that the hallucination benefit of hybrid retrieval is specific to the pre-generation retrieval quality dimension and does not reflect a difference in how faithfully the model grounds to whatever context it receives. This distinction is important for system design: both retrieval strategy and generation-stage controls (such as citation grounding) are necessary; neither is sufficient alone.

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6.2.2 Conclusion 2: Chunking Strategy is the Dominant Determinant of Context Precision

The second major conclusion is that chunking strategy exerts the largest single main effect on context precision in the experimental space ($\eta^2p = 0.915$), substantially exceeding the main effect of retrieval strategy ($\eta^2p = 0.372$). Semantic chunking (C3) consistently achieves the highest context precision across all retrieval strategy conditions, followed by C2_fixed_512, while sentence-window chunking (C4) underperforms despite its theoretical contextual expansion advantage. This conclusion directly addresses RQ2 and is the study's most structurally significant finding for pipeline design practice, because it establishes that the engineering effort invested in chunking strategy selection generates a higher return in retrieval quality than equivalent effort invested in retrieval algorithm selection.

A practically important interaction between retrieval strategy and chunking strategy was also detected ($F(6,84) = 2.28$, $p = 0.043$, $\eta^2p = 0.140$), indicating that the precision advantage of semantic chunking is partially attenuated under BM25 retrieval compared to dense or hybrid retrieval. This interaction means that chunking strategy selection should be made jointly with retrieval strategy selection, not independently.

6.2.3 Conclusion 3: Increasing Retrieval Depth Beyond top-5 Yields Meaningful Hallucination Reductions

The third conclusion is that the commonly assumed diminishing returns threshold of top-5 does not hold for the BFSI policy QA task studied here. The marginal improvement in hallucination rate from top-5 to top-10 ($\Delta = 0.040$) exceeds the pre-specified threshold of 0.02 percentage points, meaning that increasing retrieval depth to top-10 remains practically worthwhile for deployments where the doubled token cost (from ~4,600 to ~8,600 tokens per query) is acceptable. For cost-sensitive deployments, top-5 represents the most cost-efficient operating point. This conclusion directly addresses RQ3 and updates the practical guidance available in the RAG literature for financial domain applications.

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6.2.4 Conclusion 4: Prompt-Level Citation Grounding is a Reliable Zero-Cost Faithfulness Improvement

The fourth and most immediately actionable conclusion is that enabling citation grounding at the prompt level produces a statistically significant improvement in faithfulness across all retrieval strategy and chunking conditions (+0.0229, $W = 34.000$, $p < 0.0001$), without any modification to the model, the retrieval system, or the document index. This conclusion directly addresses RQ4 and constitutes the most practically actionable finding of the study: financial institutions operating RAG systems in inference-only mode can improve faithfulness by implementing citation attribution instructions in their generation prompts at zero additional computational cost.

6.2.5 Conclusion 5: The Optimal Pipeline Configuration for BFSI Policy QA

Synthesising findings across all four research questions, the study concludes that the optimal pipeline configuration for BFSI policy question answering under the compound objective of minimising hallucination rate while maximising faithfulness at moderate token cost is: R3_hybrid retrieval + C2_fixed_512 chunking + top-5 retrieval depth + G1 citation grounding. This configuration achieves a hallucination rate of 0.34, a faithfulness score of 0.985, an answer F1 of 0.089, and a mean token cost of 4,655 per query, yielding a composite score of 0.5387 the highest in the experimental space. Where token cost is unconstrained, increasing to top-10 retrieval depth (rank-2 configuration) achieves a further reduction in hallucination rate to 0.31 and near-perfect faithfulness of 0.998.

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6.3 Mapping Conclusions to Objectives

Table 6.1: Mapping of Study Conclusions to Objectives and Research Questions

Objective	**Research Question Addressed**	**Conclusion**	**Status**
Objective 1: Literature review establishing gaps	—	RAG in BFSI lacks factorial empirical comparison; gaps identified in Ch. 2	Achieved
Objective 2: Implement modular RAG pipeline	—	Full factorial pipeline implemented across 96 conditions; 14,400 observations collected	Achieved
Objective 3: Conduct factorial experiments	RQ1, RQ2, RQ3, RQ4	All four RQs answered with pre-specified statistical tests; effect sizes reported	Achieved
Objective 4: Ranked pipeline configurations	RQ1–RQ4 combined	Top-10 configurations ranked; trade-off landscape characterised	Achieved
Objective 5: Open-source release	—	Complete codebase and results CSV released at GitHub repository	Achieved

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6.4 Recommendations

6.4.1 Recommendations for BFSI Practitioners

The following recommendations are addressed to financial institutions, FinTech companies, and regulatory technology vendors responsible for designing and deploying RAG systems for policy question answering and compliance applications.

- Adopt hybrid RRF retrieval as the default retrieval strategy for financial document corpora. The 7.7 percentage-point reduction in hallucination rate compared to dense retrieval, achieved with no additional infrastructure complexity beyond maintaining a BM25 index alongside a dense vector index, represents a straightforward improvement that should be implemented as a baseline in any BFSI RAG deployment.
- Prioritise chunking strategy selection as a primary pipeline design decision, not a preprocessing afterthought. Semantic chunking (C3) or fixed-size chunking at 512 tokens (C2) should be preferred over sentence-window chunking. The engineering investment in implementing a semantic chunker using LlamaIndex's SemanticSplitterNodeParser returns substantially larger context precision improvements than equivalent investment in retrieval algorithm tuning.
- Set retrieval depth to top-5 as the default operating point and evaluate whether the available token budget permits top-10. For overnight batch processing workflows without latency constraints, top-10 provides meaningfully better hallucination suppression (HR 0.31 vs. 0.34 at top-5) at approximately double the token cost.
- Enable citation grounding in all production generation prompts as an immediate, zero-cost intervention. The prompt instruction should require the model to attribute each factual claim to a specific retrieved passage using a source identifier, and should be implemented regardless of the underlying retrieval configuration.
- For high-stakes compliance QA tasks where hallucination rates below 0.35 are a regulatory or risk management requirement, the rank-1 configuration (R3_hybrid, C2_fixed_512, top-5, G1_enabled, HR = 0.34) is recommended as the minimum sufficient configuration. Where rates below 0.32 are required, the rank-2 configuration (top-10) should be adopted.

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6.4.2 Recommendations for Future Research

The findings of this study open several productive directions for future investigation.

- Replication with larger LLM backbones. The study used LLaMA 3.2-3B as the generation backbone. Future work should replicate the factorial design with 7B, 13B, and 70B models to characterise how the interaction between retrieval quality and model scale affects the hallucination–faithfulness trade-off. The hypothesis that larger models are less sensitive to retrieval depth (having stronger parametric recall) versus more sensitive (having better utilisation of diverse retrieved contexts) is empirically unresolved in the BFSI domain.
- Extension to a larger FinanceBench corpus and additional financial document types. FinanceBench's 150 questions, while sufficient for the present study, limits the generalisability of the findings to the question types and document structures represented. Future work should evaluate on larger financial QA corpora including regulatory filings beyond SEC documents, policy manuals, and internal compliance documents.
- Evaluation of re-ranking as an additional pipeline stage. The current study does not include a cross-encoder re-ranking stage between retrieval and generation. Future work should extend the factorial design to include re-ranking e.g. with a domain-adapted cross-encoder trained on financial text as an additional independent variable, quantifying the incremental hallucination reduction achievable through re-ranking on top of RRF retrieval.
- Investigation of graph-augmented retrieval for BFSI policy QA. GraphRAG and related knowledge-graph augmentation approaches, which were explicitly excluded from this study's scope, represent a distinct architectural class with theoretical advantages for multi-hop reasoning tasks. A dedicated comparative study that evaluates GraphRAG against the baseline RAG configurations identified here as optimal would address a gap left by this study's scope constraints.
- Longitudinal assessment of citation grounding effectiveness as LLM capabilities improve. The +0.0229 faithfulness improvement from citation grounding observed in this study is specific to LLaMA 3.2-3B. As generation models improve in instruction following, the marginal benefit of citation grounding prompts may decline (if models natively generate more faithful outputs) or increase (if larger models more precisely

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implement the attribution instruction). Longitudinal or cross-model replication of the RQ4 analysis would characterise this dynamic.

6.4.3 Recommendations for Regulators and Policymakers

The findings of this study have implications for the regulatory oversight of AI systems deployed in BFSI contexts.

- Hallucination rate benchmarks should be incorporated into AI risk assessments for BFSI applications. Regulators including the FCA, ESMA, and OCC should consider requiring financial institutions deploying LLM-based advisory or compliance systems to report hallucination rates on standardised benchmarks as part of model risk management submissions. The experimental methodology of this study provides a replicable framework for such measurement.
- Open benchmarks for financial AI evaluation should be supported and extended. FinanceBench's grounded, publicly available question-answer pairs with authoritative document provenance provide an ideal basis for regulatory benchmarking. Supporting the development of larger, more diverse financial QA benchmarks, including multi-document, cross-referencing, and temporally dynamic question types, would strengthen the evaluation infrastructure available to both regulators and practitioners.
- RAG architecture adoption should be actively encouraged as a hallucination mitigation mechanism. The results of this study demonstrate that well-configured RAG systems achieve hallucination rates below 0.35 on challenging financial document QA tasks. Regulatory guidance that explicitly recognises RAG as an approved architecture for hallucination mitigation, conditional on the adoption of recommended pipeline configurations, would lower barriers to AI deployment in compliance-sensitive workflows.

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6.5 Contribution to Knowledge

This study makes five original contributions to knowledge at the intersection of LLM hallucination mitigation, RAG pipeline engineering, and financial domain NLP.

- First, it provides the first published factorial empirical comparison of dense, sparse, and hybrid retrieval strategies on the FinanceBench benchmark with hallucination rate and faithfulness as primary outcomes. The factorial design enables the quantification of main effects and interaction effects that single-variable studies cannot detect.
- Second, it establishes the dominant role of chunking strategy as the primary determinant of context precision in BFSI RAG pipelines, with an effect size ($\eta^2 p = 0.915$) that substantially exceeds the effect of retrieval strategy selection. This finding reorients the practitioner attention hierarchy for RAG pipeline optimisation.
- Third, it extends the diminishing returns analysis for retrieval depth to the BFSI domain, demonstrating that meaningful hallucination reductions persist to top-10 retrieval depth for financial document QA, contradicting the common top-5 heuristic.
- Fourth, it provides the first empirical evidence that prompt-level citation grounding, without model fine-tuning, produces a statistically significant and consistent faithfulness improvement in BFSI policy QA, making this a practically actionable finding for the largest class of production RAG deployments.
- Fifth, it releases a complete open-source implementation of the experimental pipeline, including all 96 configuration variants, evaluation scripts, and a full results dataset, enabling independent replication and extension of the study's findings.

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6.6 Reflections on the Study's Broader Significance

Financial institutions that deploy more reliable AI systems provide better outcomes for the customers, businesses, and communities they serve. Compliance failures triggered by AI-generated misinformation impose costs that manifest in regulatory fines, litigation, reputational damage and most importantly harm to individuals who rely on financial institutions to provide accurate, trustworthy guidance. The finding that well-configured hybrid RAG pipelines with semantic or 512-token chunking, top-5 retrieval depth, and citation grounding can achieve hallucination rates as low as 0.31 on the FinanceBench benchmark represents meaningful empirical progress toward RAG systems that are reliable enough for deployment in compliance-critical workflows.

The study does not claim that these configurations are ready for deployment without human oversight; a hallucination rate of 0.31 still implies that nearly one in three responses contains at least one unsupported factual claim, which is an unacceptable error rate for high-stakes regulatory applications without additional safeguards. However, it establishes that structured, evidence-based pipeline design can reduce hallucination rates substantially relative to naively configured RAG systems, and that the design choices required to achieve these improvements are accessible to any organisation with a standard cloud GPU inference environment.

The release of the complete codebase and experimental results representing over 14,000 pipeline executions and 96 factorial conditions ensures that the evidence base constructed by this study is not locked behind a single publication, but is available for the broader research community, BFSI practitioners, and regulatory bodies to inspect, reproduce, and extend. In this sense, the study's contribution to trustworthy AI in financial services extends beyond its findings to its commitment to open, reproducible science.

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6.7 Final Conclusion

This thesis set out to answer a specific and consequential empirical question: how should a RAG pipeline be configured to minimise hallucination and maximise faithfulness in BFSI policy question answering? The answer, supported by a controlled factorial experiment across 96 pipeline configurations and 14,400 query-level observations, is clear. Hybrid RRF retrieval, combined with semantic or 512-token fixed-size chunking, a retrieval depth of top-5 or top-10, and prompt-level citation grounding, constitutes the evidence-based optimal configuration for this task. These findings are novel, domain-specific, statistically robust, and immediately actionable. The financial services industry deserves AI systems that ground their outputs in authoritative evidence rather than statistical plausibility; this study provides the empirical foundation upon which such systems can be responsibly built.

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Appendix A
Research Proposal & Code Repository

A.1 Research Proposal

The full approved research proposal submitted in March 2026, covering the study background, research questions, methodology, experimental design, and project plan, is available at:

https://drive.google.com/file/d/1SudqzDtU-NW6FZPOknW-XRH8yHA0c874/view?usp=sharing

A.2 Open-Source Code Repository

In accordance with the reproducibility commitment stated in Section 5.1 and Objective 5 of this study, all pipeline code, configuration files, evaluation scripts, and experimental results have been made publicly available under an open-source licence. The repository contains everything required to independently replicate all 96 pipeline configurations and 14,400 query-level observations reported in this thesis.

Repository URL:

https://github.com/andysharma1997/shubham_sharma_mitigation_of_hallucination_in_llms_using_rag

Full Factorial Results (CSV):

https://github.com/andysharma1997/shubham_sharma_mitigation_of_hallucination_in_llms_using_rag/blob/main/final_results.csv

The repository is hosted on GitHub under the account andysharma1997 and is set to public visibility. Any researcher, practitioner, or reviewer may clone, fork, or download the repository without restriction. Issues, questions, and pull requests are welcomed via the GitHub interface.

A.3 Experimental Results Data

The file **final_results.csv** contains the complete results of all 96 pipeline configurations evaluated across 150 FinanceBench questions. Each row records the following fields as the mean value across all 150 questions for that configuration:

Table A.1: Schema for final_results.csv — 96 rows × 14 columns

Column Name	Type	Description

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retrieval_strategy	String	Retrieval paradigm: dense vector search, sparse BM25, or hybrid RRF
chunking_strategy	String	Document segmentation method: fixed 256-token, fixed 512-token, semantic, or sentence-window
top_k	Integer	Number of passages retrieved and passed to the generation context per query
citation_grounding	String	Whether the generation prompt requires explicit claim-level source attribution
hallucination_rate	Float	Mean proportion of responses containing ≥1 unsupported factual claim, measured via DeepEval G-Eval
faithfulness	Float	RAGAS faithfulness: mean fraction of answer claims directly attributable to retrieved passages
context_precision	Float	RAGAS context precision: proportion of retrieved chunks that are relevant to the query
context_recall	Float	RAGAS context recall: proportion of ground-truth answer content covered by retrieved chunks
answer_f1	Float	Token-level F1 score against FinanceBench ground-truth answers
answer_em	Integer	Exact match against ground-truth answer (1 = exact match, 0 = no match); all configurations scored 0 on this corpus

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latency_mean_s	Float	Mean wall-clock time (seconds) from query submission to full response receipt across 150 queries
latency_p95_s	Float	95th-percentile latency (seconds) — captures tail latency behaviour under the 150-query evaluation
token_cost_mean	Float	Mean total tokens (prompt + completion) consumed per query; scales with top-k and chunk size
index_time_seconds	Float	Time in seconds to build the document index for the given chunking configuration (per-config, not per-query)

A.4 How to Cite This Repository

If using this repository or the experimental results in academic work, please cite as:

> Sharma, S. (2026). *Mitigating Hallucinations in Large Language Models Using Retrieval-Augmented Generation* [Source code]. GitHub. https://github.com/andysharma1997/shubham_sharma_mitigation_of_hallucination_in_llms_using_rag

A.5 Licence and Reuse

The repository is released under the MIT Licence, permitting unrestricted reuse, modification, and redistribution for academic, commercial, and non-commercial purposes, provided the original authorship is acknowledged. The FinanceBench dataset is subject to its own licence terms as specified by Islam et al. (2023); users should refer to the FinanceBench repository directly for data access and usage rights.

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Appendix B

Research Ethics Review Form

B.1 Project Details

Date of Submission: Jun 2026

Student Name Shubham Sharma	**Student ID** 1197070
Programme MSc Machine Learning & Artificial Intelligence	**Academic Year** 2025–2026
Supervisor Dr. Sitanath Biswas	**Delivery Partner** upGrad

Project Title
Mitigating Hallucinations in Large Language Models Using Retrieval-Augmented Generation

B.2: PROJECT SUMMARY

Provide a brief, plain-English summary of the research project, including its aims, methods, and expected outputs.

This study conducts a factorial empirical evaluation of Retrieval-Augmented Generation (RAG) pipeline configurations to identify which combinations of retrieval strategy, chunking approach, retrieval depth, and citation grounding most effectively reduce hallucinations in Banking, Financial Services, and Insurance (BFSI) policy question answering. The study uses the publicly available FinanceBench benchmark dataset (150 questions grounded in real SEC filings). All experimentation is conducted computationally using open-source LLMs (LLaMA 3.2-3B) and automated evaluation frameworks (RAGAS, DeepEval). No human participants, personal data, or sensitive information are involved.

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B.3: ETHICAL RISK SCREENING CHECKLIST

Please answer all questions. If you answer YES to any question, provide justification in Section B.4.

Does this research involve human participants (interviews, surveys, observations)?	Yes ☐	No ☑
Does this research involve children or vulnerable adults?	Yes ☐	No ☑
Does this research involve collection, processing, or storage of personal data?	Yes ☐	No ☑
Could any aspect of this research cause physical, psychological, or reputational harm to participants?	Yes ☐	No ☑
Does this research involve deception of participants?	Yes ☐	No ☑
Does this research involve covert observation?	Yes ☐	No ☑
Does this research involve animals?	Yes ☐	No ☑
Does this research involve access to sensitive, confidential, or classified information?	Yes ☐	No ☑
Does this research involve dual-use technologies that could be misused for harmful purposes?	Yes ☐	No ☑
Does this research involve financial incentives for participants?	Yes ☐	No ☑
Does this research require access to commercially proprietary datasets or models?	Yes ☐	No ☑
Could the findings of this research have adverse societal, environmental, or legal implications?	Yes ☐	No ☑
Does this research involve the use of generative AI systems in a way that could produce harmful outputs?	Yes ☐	No ☑
Does this research collect data via online platforms, social media, or web scraping?	Yes ☐	No ☑
Does this research require ethics approval from an external body (NHS, MoD, corporate partner, etc.)?	Yes ☐	No ☑

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B.4: JUSTIFICATION FOR ANY 'YES' RESPONSES

If you answered YES to any question in Section B.3, provide a full justification and describe the safeguards in place.

Not applicable. All questions in Section 3 were answered No. This research is entirely computational in nature, using publicly available benchmark datasets and open-source models. No human participants, personal data, or sensitive material are involved at any stage.

B.5: DATA MANAGEMENT

What data will be collected?	Automated evaluation scores (hallucination rate, faithfulness, context precision, answer F1, token cost, latency) generated by running publicly available LLMs and evaluation frameworks against the FinanceBench dataset.
Does the data contain personal information?	No. All data is system-generated metric output. No personal or identifiable information is collected.
Where will data be stored?	Locally on a secured cloud GPU instance (AWS g4dn.xlarge) and in a public GitHub repository. No institutional data repositories containing sensitive data are used.
Who will have access to the data?	The researcher (Shubham Sharma), the supervisor (Dr. Sitanath Biswas), and the general public via the open-source GitHub repository.
How long will data be retained?	For the duration of the research and for a minimum of five years thereafter, in line with LJMU's research data retention policy.
Will data be shared or published?	Yes. All results will be published openly in the GitHub repository and reported in the thesis. No data requires restriction.
Is a Data Protection Impact Assessment required?	No. No personal data is processed. The UK GDPR does not apply to this research.

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B.6: INTELLECTUAL PROPERTY AND OUTPUTS

Will outputs be published?	Yes as an MSc dissertation submitted to LJMU, and as an open-source codebase on GitHub.
Ownership of outputs	Intellectual property resides with the student (Shubham Sharma) and LJMU in accordance with the University's IP Policy.
Open-source licence	MIT Licence permitting unrestricted academic and commercial reuse with attribution.
Use of third-party datasets	FinanceBench (Islam et al., 2023) is used under its publicly available licence. No proprietary datasets are used.
Use of open-source models	LLaMA 3.2-3B is used under the Meta LLaMA 3 Community Licence. All other tools are used under Apache 2.0 or MIT licences.
Commercial exploitation intended?	No.

B.7: RESEARCHER DECLARATION

I confirm that the information provided in this form is accurate and complete to the best of my knowledge. I understand my responsibilities under the LJMU Code of Practice for Research Ethics, the UK General Data Protection Regulation (UK GDPR), and the University's Research Integrity Policy. I confirm that this research will be conducted in accordance with the ethical principles outlined above and that any significant changes to the research design will be reported to my supervisor and, where necessary, referred back for further ethical review.

SHUBHAM SHARMA	29 Jun 2026
Researcher (Shubham Sharma) Signature	*Date*

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B.8: ETHICS RISK CATEGORY (FOR OFFICE USE)

☑ **Category A** *Low Risk — Supervisor Approval Sufficient*	☐ Category B *Medium Risk — Faculty Ethics Committee Review*	☐ Category C *High Risk — Full University Ethics Review Required*

This research is classified as Category A (Low Risk). It is entirely computational, uses only publicly available datasets and open-source tools, involves no human participants or personal data, and presents no identified dual-use or societal harm risks. Supervisor approval is sufficient; Faculty Ethics Committee referral is not required.